

Isotopie

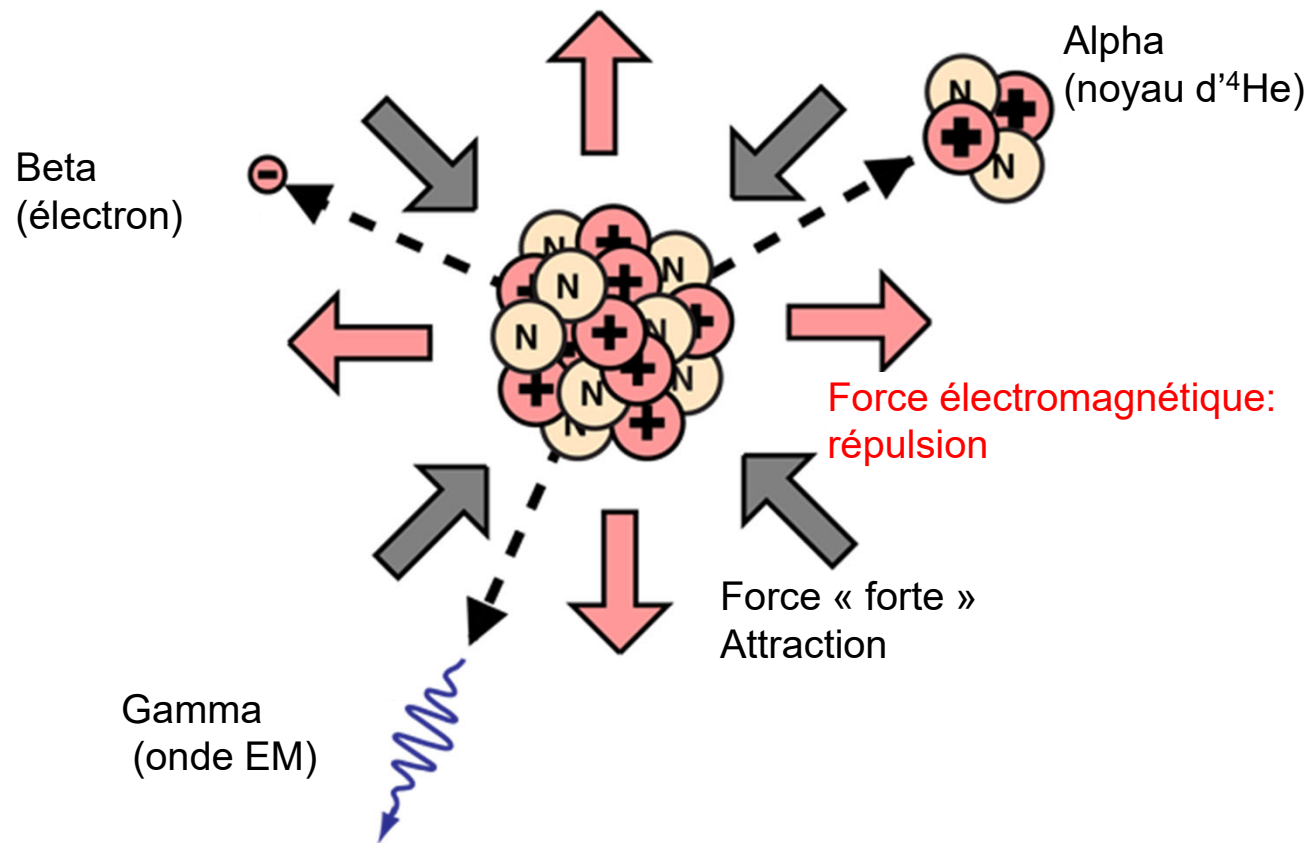
CHM-2150

Chimie de l'environnement

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Hiver 2024
VCH-2250-C
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Isotopes stables vs radioisotopes



[illegible]

Carte des nucléides: isotopes (protons)

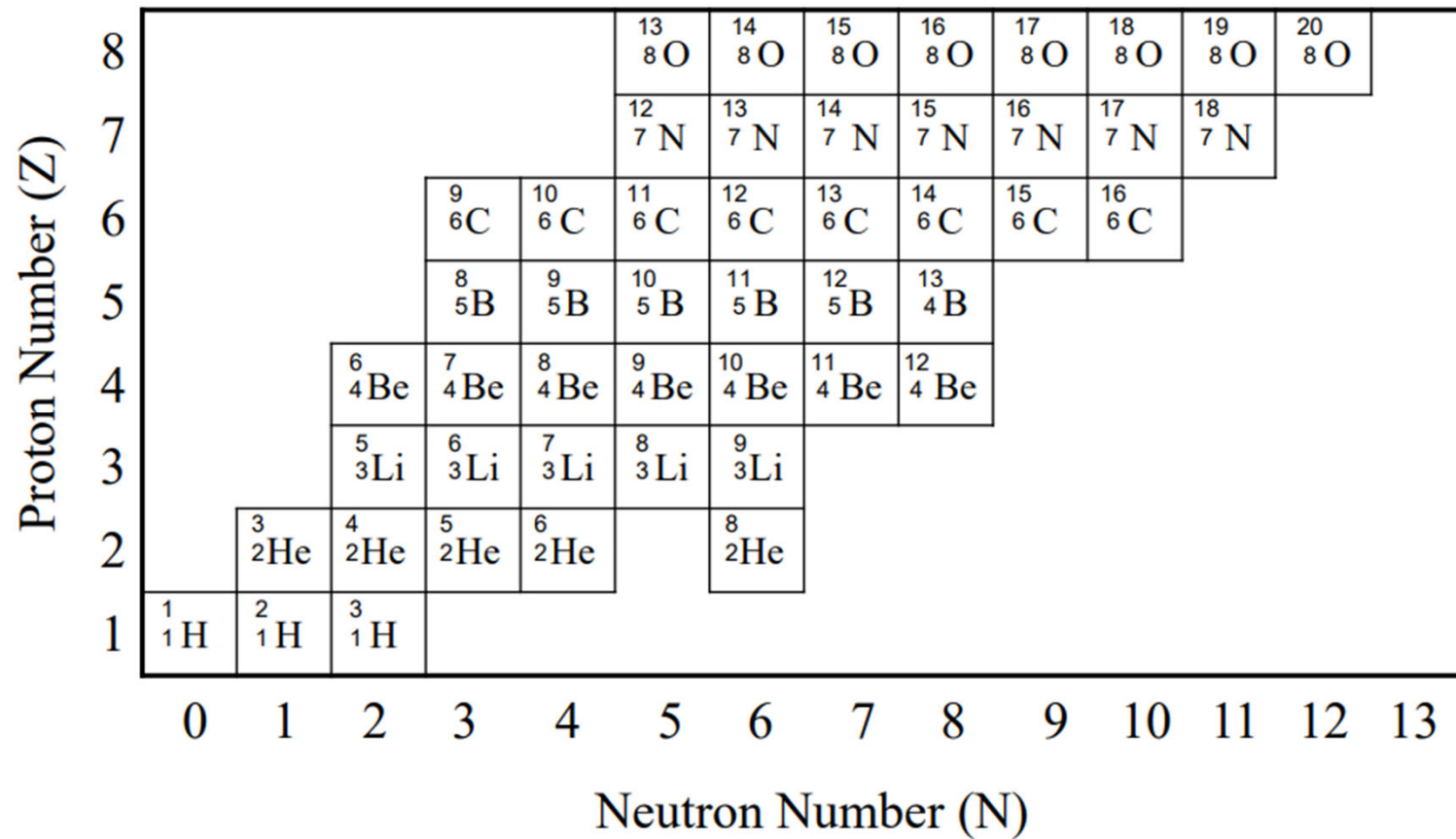
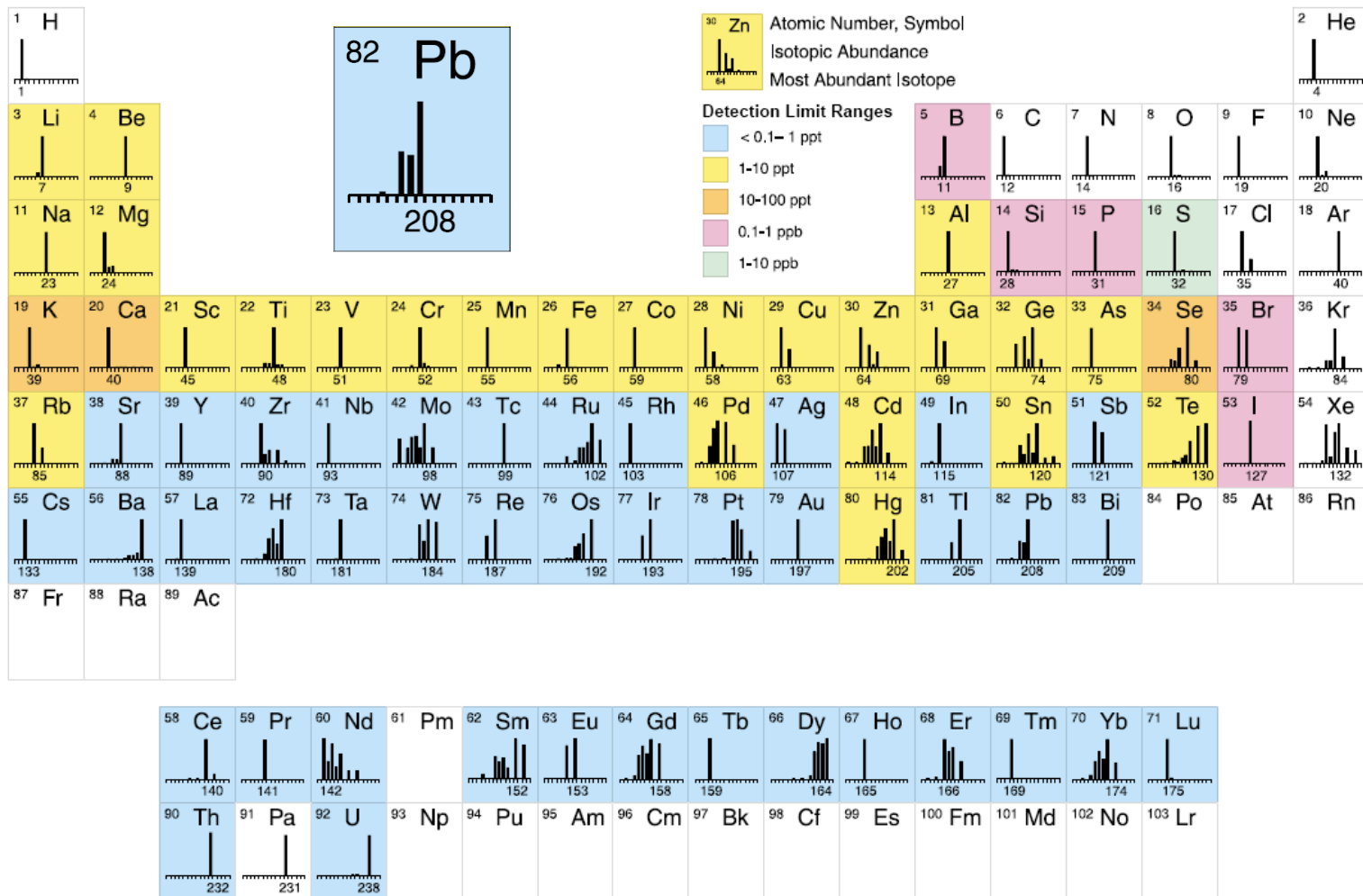


Tableau périodique et isotopes stables

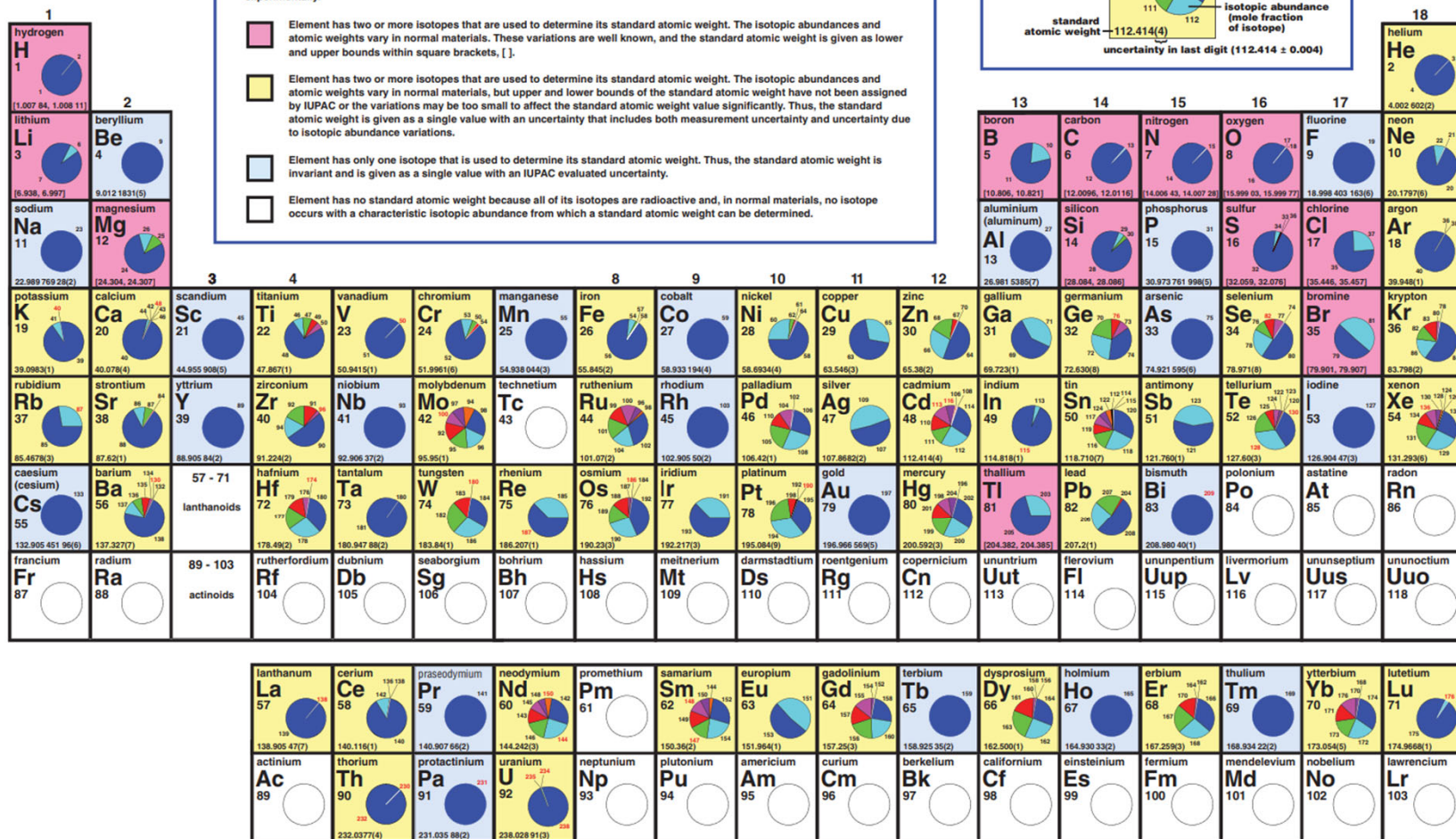
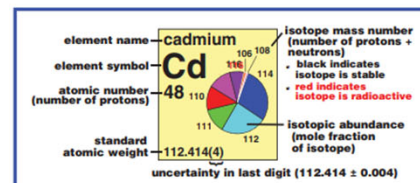


IUPAC Periodic Table of the Isotopes

Element Background Color Key

Standard atomic weights are the best estimates by IUPAC of atomic weights that are found in normal materials, which are terrestrial materials that are reasonably possible sources for elements and their compounds in commerce, industry, or science. They are determined using all stable isotopes and selected radioactive isotopes (having relatively long half-lives and characteristic terrestrial isotopic compositions). Isotopes are considered stable (non-radioactive) if evidence for radioactive decay has not been detected experimentally.

-  Element has two or more isotopes that are used to determine its standard atomic weight. The isotopic abundances and atomic weights vary in normal materials. These variations are well known, and the standard atomic weight is given as lower and upper bounds within square brackets, [].
-  Element has two or more isotopes that are used to determine its standard atomic weight. The isotopic abundances and atomic weights vary in normal materials, but upper and lower bounds of the standard atomic weight have not been assigned by IUPAC or the variations may be too small to affect the standard atomic weight value significantly. Thus, the standard atomic weight is given as a single value with an uncertainty that includes both measurement uncertainty and uncertainty due to isotopic abundance variations.
-  Element has only one isotope that is used to determine its standard atomic weight. Thus, the standard atomic weight is invariant and is given as a single value with an IUPAC evaluated uncertainty.
-  Element has no standard atomic weight because all of its isotopes are radioactive and, in normal materials, no isotope occurs with a characteristic isotopic abundance from which a standard atomic weight can be determined.



Pourquoi mesurer les isotopes ?

Datation :

les isotopes issus de la désintégration d'élément radioactif

Source de contaminants:

origine, transfert et mélange de minéral

Paléothermomètres:

températures de l'océan, de l'atmosphère, de cristallisation

Géochimie:

mécanismes responsables de la mobilité des éléments chimiques

Notation: ratio, delta et facteur de fractionnement

Composition isotopique

$$R_A = \frac{\text{quantité de H}_2^{18}\text{O}}{\text{quantité de H}_2^{16}\text{O}}$$

Facteur de fractionnement

Normalement, l'isotope le plus lourd est utilisé au numérateur

$$\alpha_{A-B} = \frac{R_A}{R_B}$$

$$\delta^{18}\text{O} = \frac{R_A - R_{\text{ref}}}{R_{\text{ref}}} \cdot 1000 = \text{‰}$$

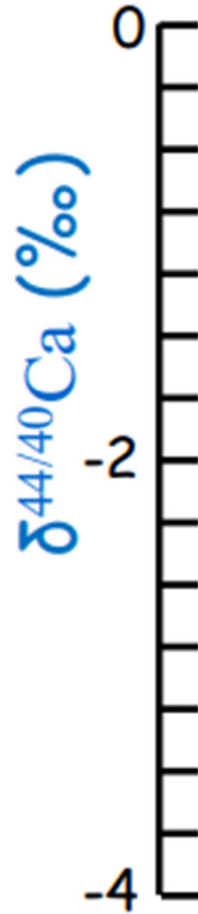
Expression en « pour mille »

Notation: ratio, delta et facteur de fractionnement

$$\delta^{44}\text{Ca} = 0 \text{ ‰}$$

$$\delta^{44}\text{Ca} > 0 \text{ ‰}$$

$$\delta^{44}\text{Ca} < 0 \text{ ‰}$$



Composition isotopique dans les matériaux certifiés

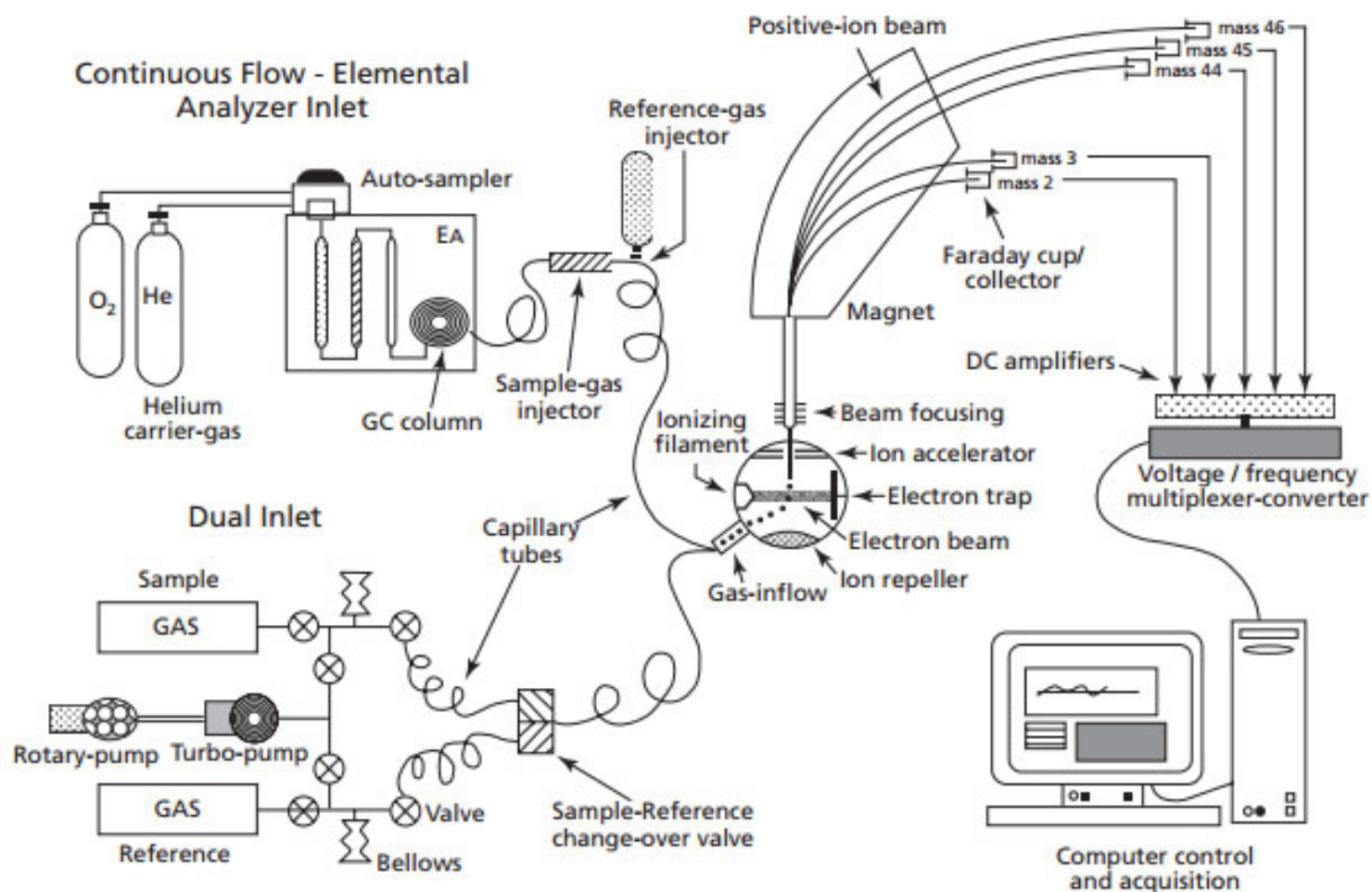
Éléments	Isotope	Abondance (%)	Différence relative de masse (%)	Standards internationaux	Abondance absolue dans standard (R_{standard})
Hydrogène	^1H	99.985	100	Vienna Standard Mean Ocean Water (VSMOW)	$^2\text{H}:^1\text{H} = 0.00015576$
	^2H (ou D)	0.0155			
Carbone	^{12}C	98.892	8.3	Vienne Pee Dee Belemnite (VPDB)	$^{13}\text{C}:^{12}\text{C} = 0.0112372$
	^{13}C	1.108			
Azote	^{14}N	99.635	7.1	Atmospheric nitrogen	$^{15}\text{N}:^{14}\text{N} = 0.0036765$
	^{15}N	0.365			
Oxygène	^{16}O	99.759	12.5 ($^{18}\text{O}:^{16}\text{O}$)	VSMOW (H ₂ O); VPDB (CO ₂ ou carbonate)	VSMOW = 0.0020052 VPDB = 0.0020672 ($^{18}\text{O}:^{16}\text{O}$)
	^{17}O	0.037			
	^{18}O	0.204			
Soufre	^{32}S	95	6.3 ($^{34}\text{S}:^{32}\text{S}$)	Vienna Canon Diablo meteorite troilite (VCDT)	$^{34}\text{S}:^{32}\text{S} = 0.0450045$
	^{33}S	0.75			
	^{34}S	4.21			
	^{36}S	0.014			

Composition isotopique dans les matériaux certifiés

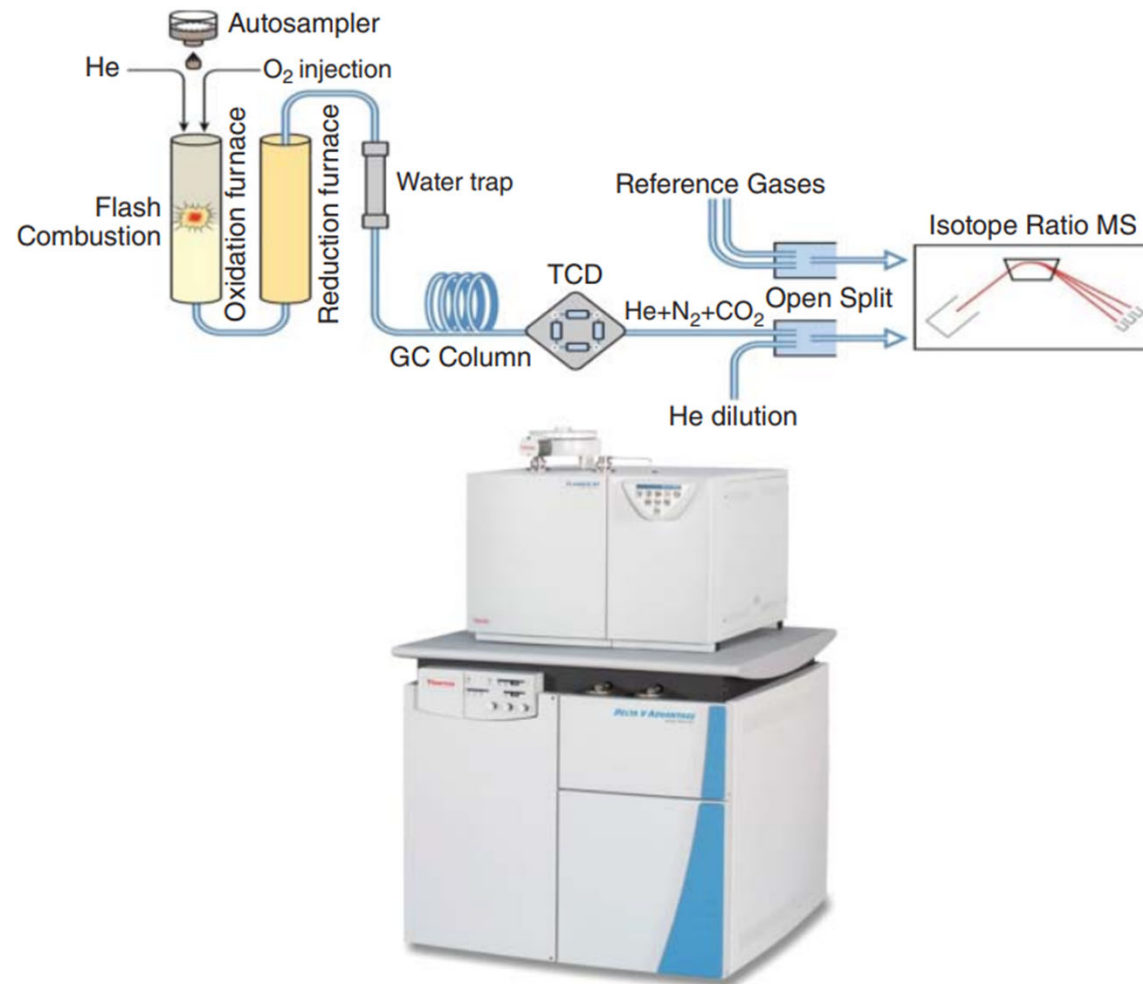
Éléments	Isotope	Abondance (%)	Différence relative de masse (%)	Standards internationaux	Abondance absolue dans standard (R_{standard})
Fer	^{54}Fe	5.82	3.7 ($^{56}\text{Fe}:$ ^{54}Fe)	Moyenne terrestre et lunaire	$^{56}\text{Fe}:$ $^{54}\text{Fe} = 15.7028$
	^{56}Fe	91.66			
	^{57}Fe	2.19			
	^{58}Fe	0.33			
Strontium	^{84}Sr	0.56	1.1 ($^{86}\text{Sr}:$ ^{87}Sr)	USGD Tridacna	$^{87}\text{Sr}:$ $^{86}\text{Sr} = 0.709249$
	^{86}Sr	9.87			
	^{87}Sr	704			
	^{88}Sr	82.53			

- Disponible pour tous en quantité suffisante
- Homogène et stable
- Possibilité de travailler avec un laboratoire certifié de niveau international

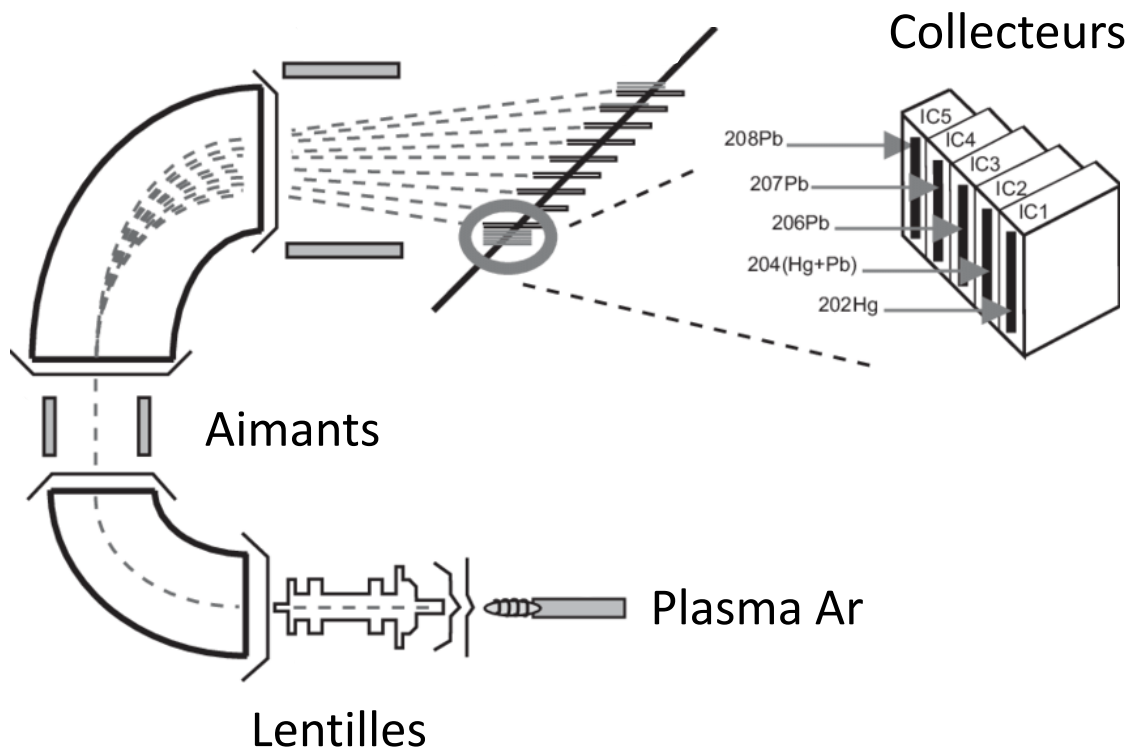
Dual inlet/continuous flow elemental analyzer



Continuous flow isotope-ratio mass spectrometer

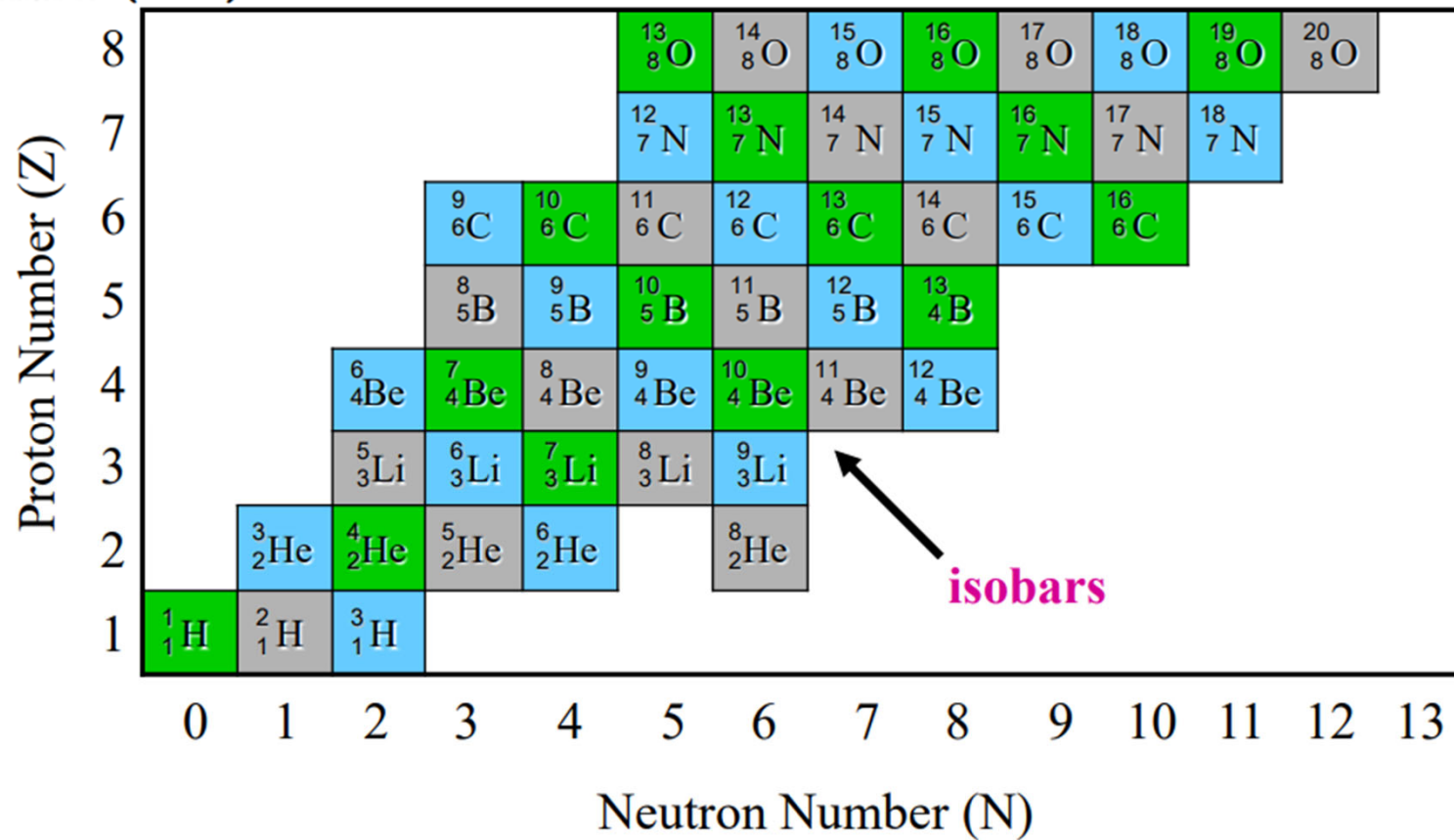


Multi-collector ICP mass spectrometer

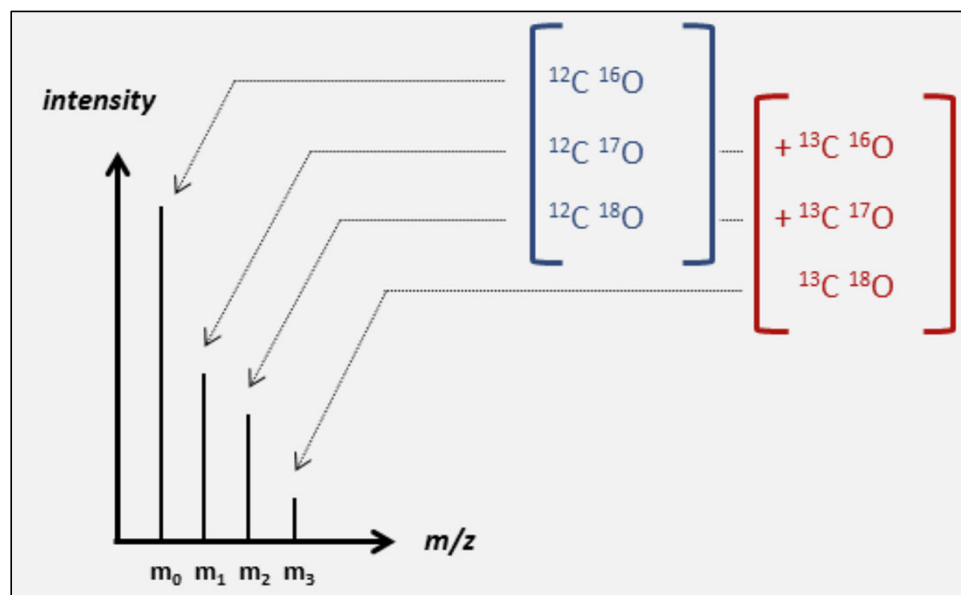


Thermo Neptune MC-ICP-MS

Interférences isobariques



Isotopologues

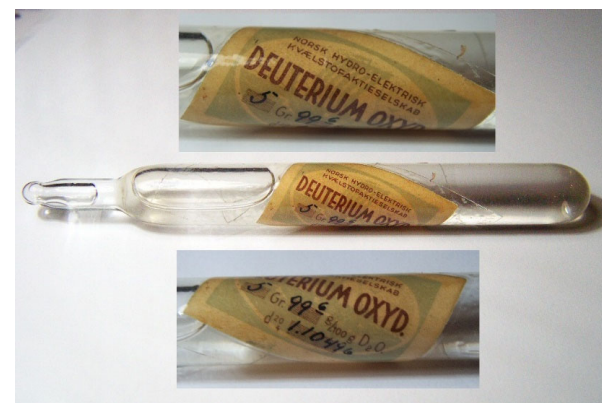


Relative abundances of the isotopologues of CO_2

Mass	Isotopologue	Relative Abundance
44	$^{12}\text{C}^{16}\text{O}_2$	98.40%
45	$^{13}\text{C}^{16}\text{O}_2$	1.11%
	$^{12}\text{C}^{17}\text{O}^{16}\text{O}$	748 ppm
46	$^{12}\text{C}^{18}\text{O}^{16}\text{O}$	0.40%
	$^{13}\text{C}^{17}\text{O}^{16}\text{O}$	8.4 ppm
	$^{12}\text{C}^{17}\text{O}_2$	0.142 ppm
47	$^{13}\text{C}^{18}\text{O}^{16}\text{O}$	44.4 ppm
	$^{12}\text{C}^{17}\text{O}^{18}\text{O}$	1.50 ppm
	$^{13}\text{C}^{17}\text{O}_2$	1.60 ppb
48	$^{12}\text{C}^{18}\text{O}_2$	3.96 ppm
	$^{13}\text{C}^{17}\text{O}^{18}\text{O}$	16.8 ppb
49	$^{13}\text{C}^{18}\text{O}_2$	44.5 ppb

Isotopologues et effets isotopiques

Propriété	H ₂ O	D ₂ O
Point de fusion (°C)	0	3.82
Point d'ébullition (°C)	100.0	101.72
Densité (20°C, g/mL)	0.9982	1.1056
Température à densité maximale (°C)	4.0	11.6
Viscosité (20°C, centipoise)	1.005	1.25
Tension de surface (25°C, dyn*cm)	71.97	71.93
p«H» (25°C)	7.00	7.41
Chaleur de fusion (cal/mol)	1 436	1 515
Chaleur de vaporisation (cal/mol)	10 515	10 864



Contenu en D dans l'eau:

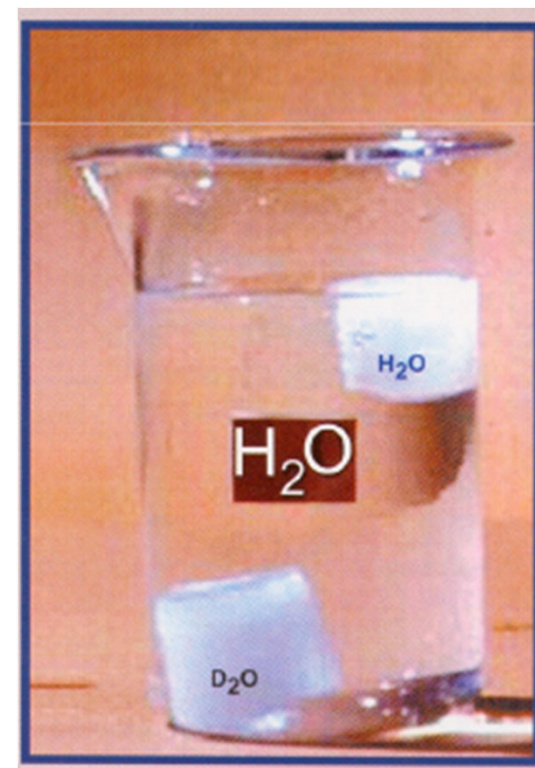
- 25% - en bonne santé, mais stérile
- 25% - 90% effet neurologique (partiellement réversible)
- >90% - mort

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Composition isotopique

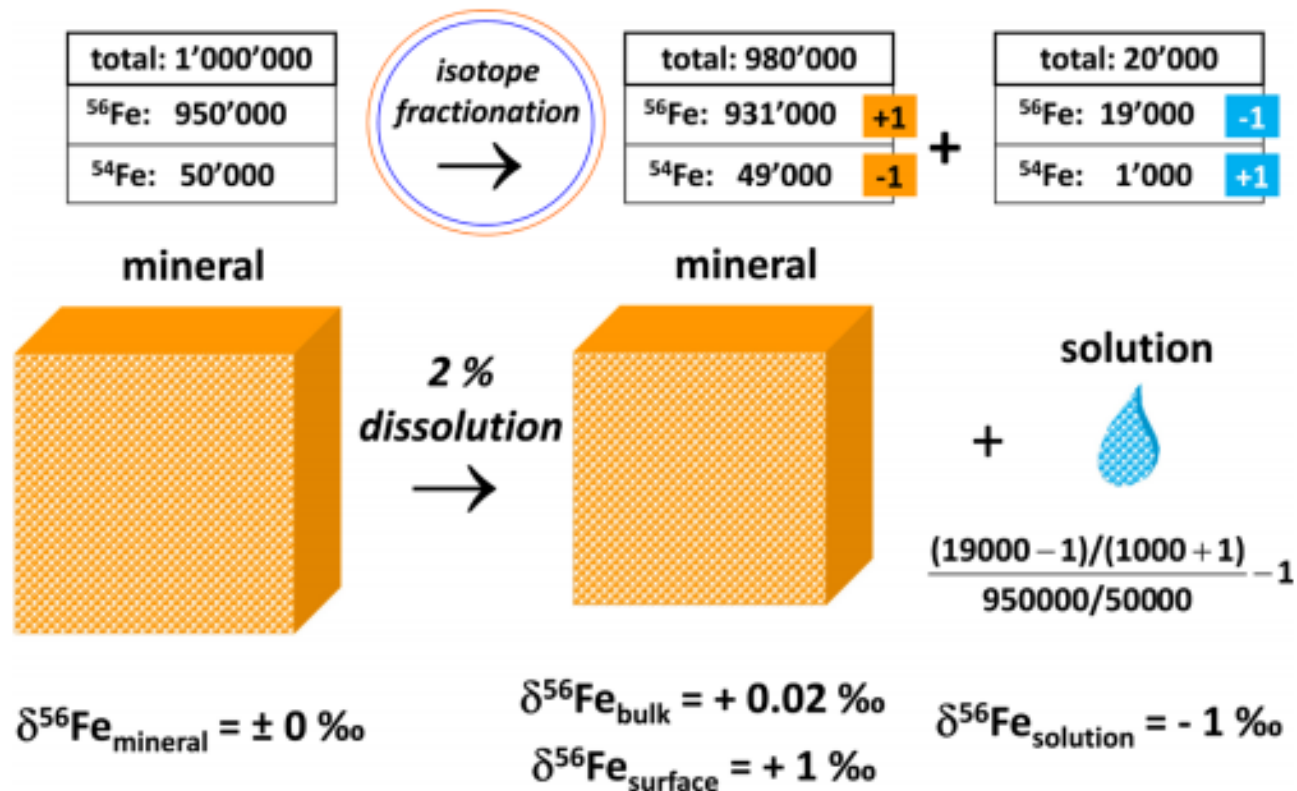
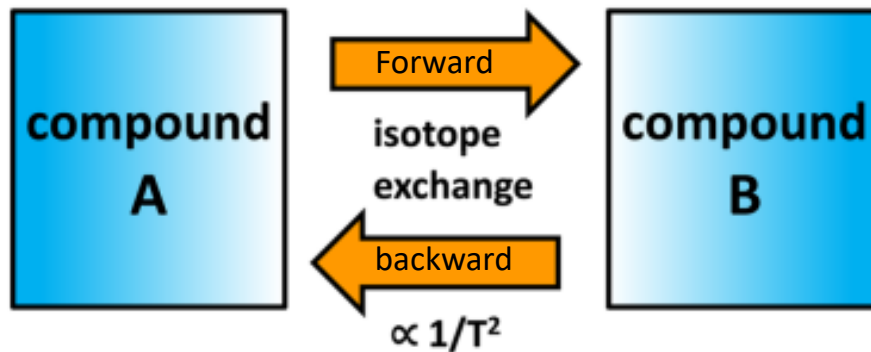


Figure 2. Schematic illustration of metal stable isotope fractionation during mineral dissolution and delta values of the involved metal pools (adapted from Wiederhold²⁹⁷). In the presented example, a small fraction (2%) of an iron mineral with 1 million Fe atoms is dissolved (simplified boundary conditions: 95% ⁵⁶Fe + 5% ⁵⁴Fe; the surface layer corresponds to 4% of the total mineral). Without isotope fractionation, the dissolved pool would contain 1000 ⁵⁴Fe atoms. However, metal isotope fractionation manifests itself by the preferential release of one light ⁵⁴Fe atom resulting in a $\delta^{56}\text{Fe}$ value of -1 ‰ in solution. The influence on the isotopic signature of the bulk mineral is negligible ($+0.02 \text{ ‰}$) due to its larger reservoir size. However, a relative enrichment of heavy isotopes is created at the mineral surface. The general magnitude of metal stable isotope fractionation illustrated in this example is similar to the extent of fractionation observed in nature. However, in order to compare the dimensions, one has to keep in mind that one gram of soil usually contains about 10^{20} Fe atoms.

Équilibre (processus bidirectionnel, p.ex. précip./diss.)

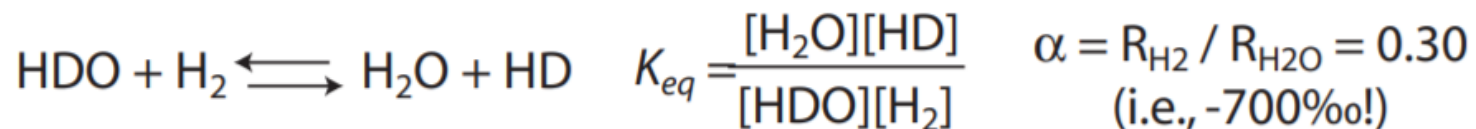
L'énergie de vibration, rotation et translation dépend des masses des atomes



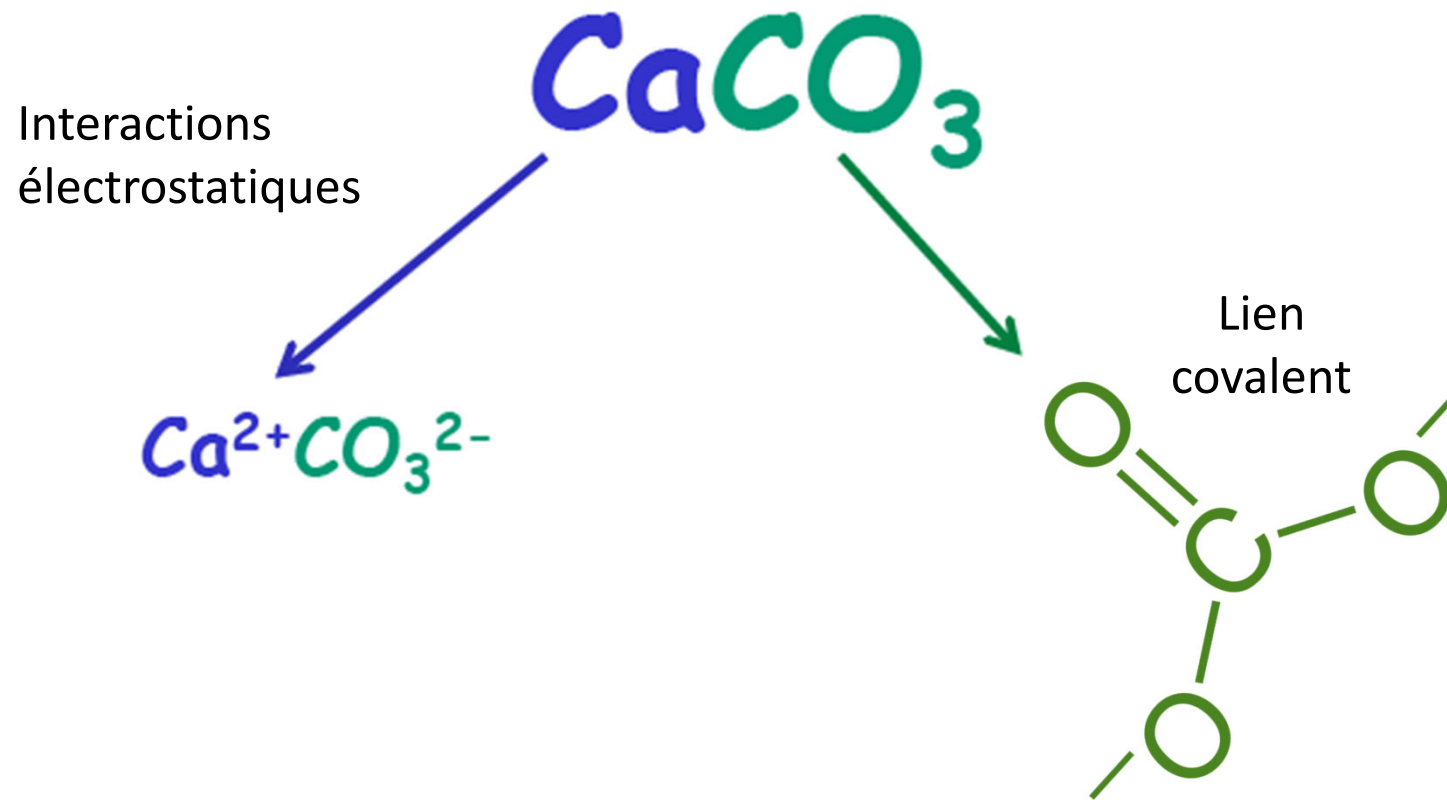
Résulte de la température

$$K_{eq} = \alpha_{A-B} = \frac{{}^2K_f / {}^1K_b}{{}^iK_f / {}^iK_b}$$

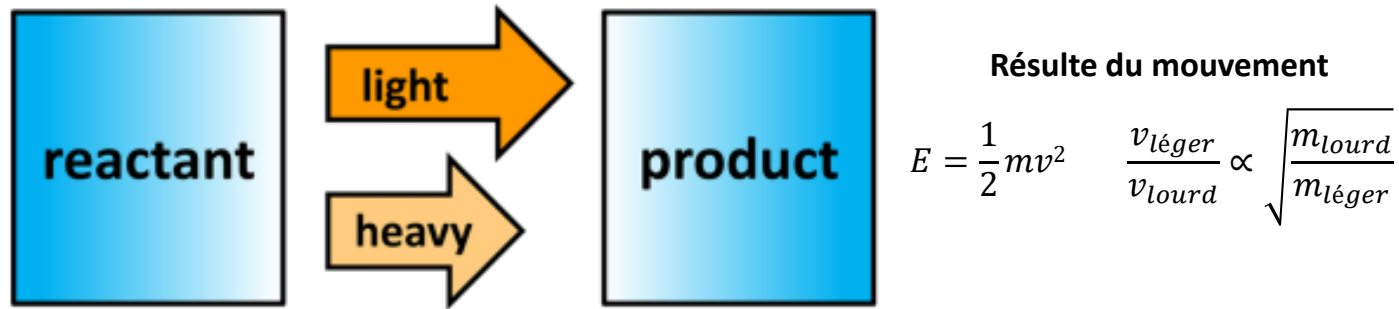
- Le rapport des isotopes dans chaque compartiment est constant à une température donnée



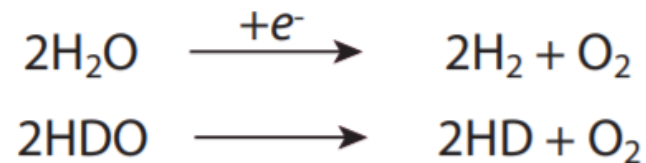
Équilibre (processus bidirectionnel, p.ex. précip./diss.)



Cinétique (processus unidirectionnel, p.ex. enzyme)



- La différence dans les vitesses de réaction des isotopologues dépend du ratio des masses des isotopes impliqués et de leur fréquence de vibration
- Processus irréversible (redox, enzymes)
- Plus le lien à une énergie faible, plus l'énergie potentielle pour le briser est faible

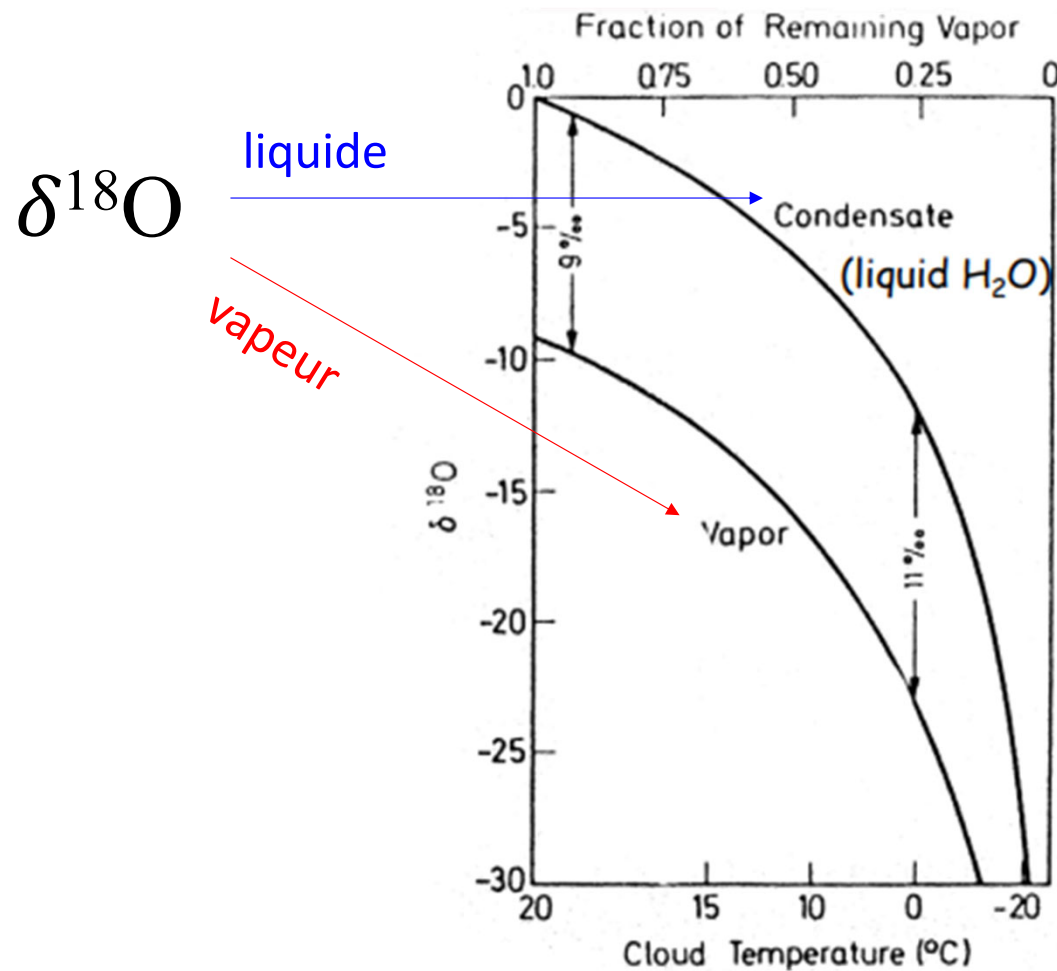


Fractionnement équilibre et cinétique: résumé

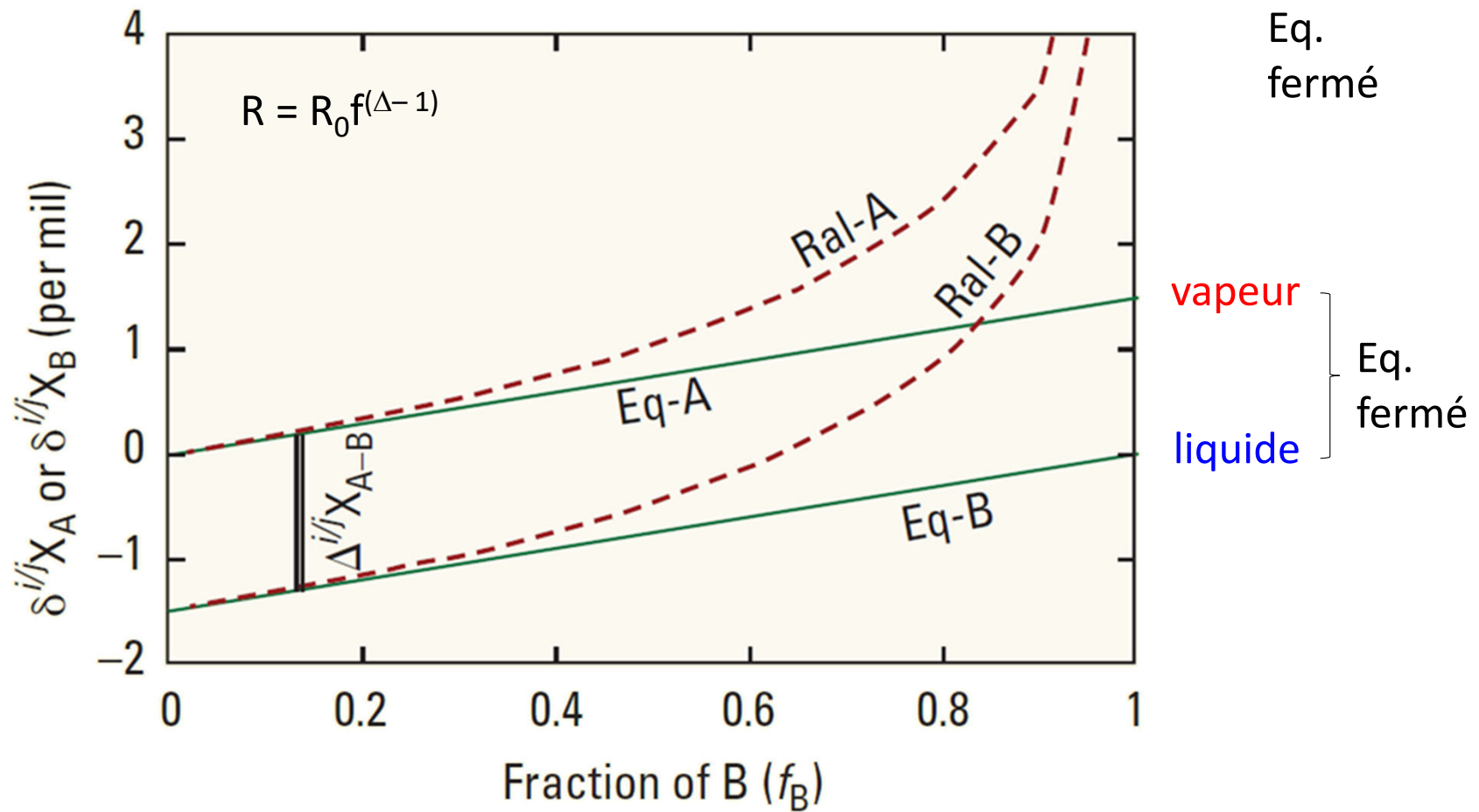
- **Équilibre** : Le rapport des isotopes dans chaque compartiment est constant à une température donnée
- **Cinétique**: Les liens avec des isotopes légers sont brisés plus rapidement dans une réaction unidirectionnelle

Que se passe-t-il lorsque les compartiments A et B sont dans un système ouvert (pas d'échange) ?

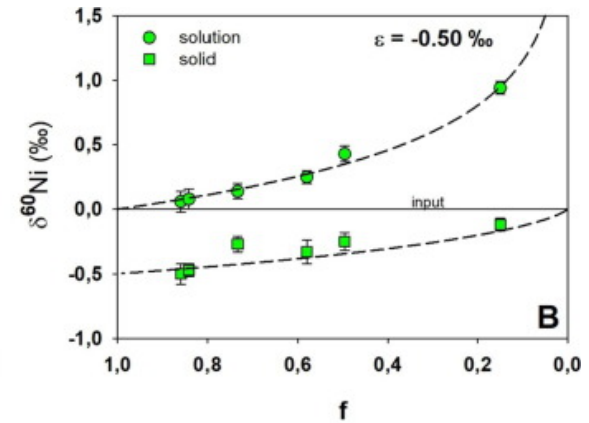
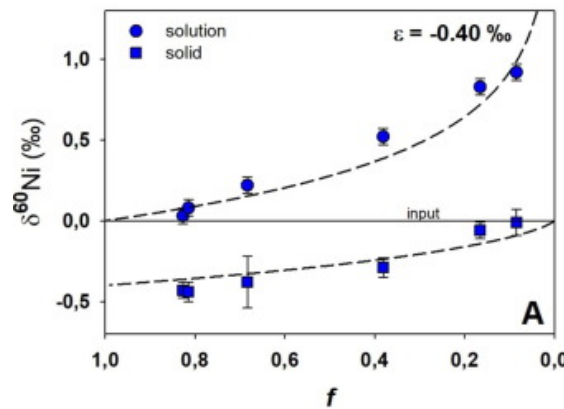
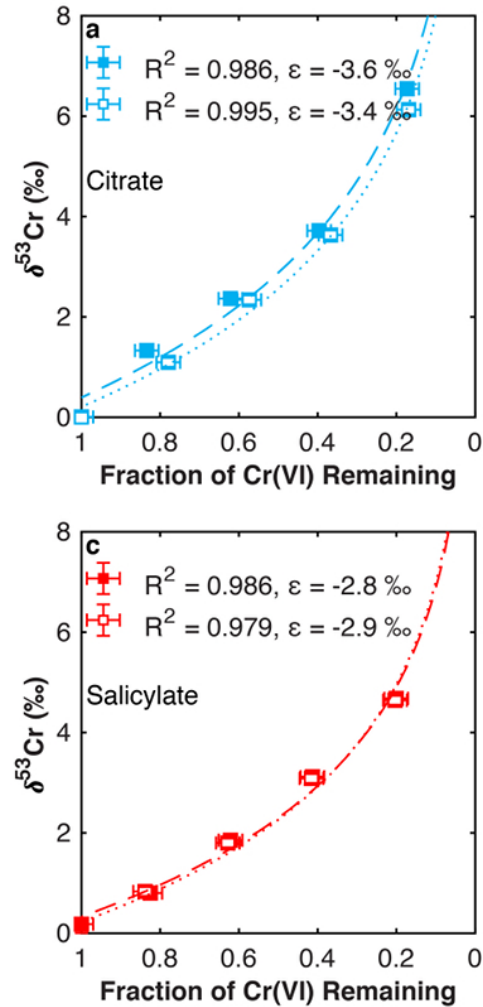
Distillation de Rayleigh : équilibres unidirectionnels



Distillation de Rayleigh : équilibres unidirectionnels



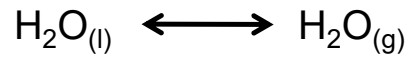
Expériences en laboratoire : réduction et précipitation



[doi:10.1016/j.gca.2021.10.027](https://doi.org/10.1016/j.gca.2021.10.027)

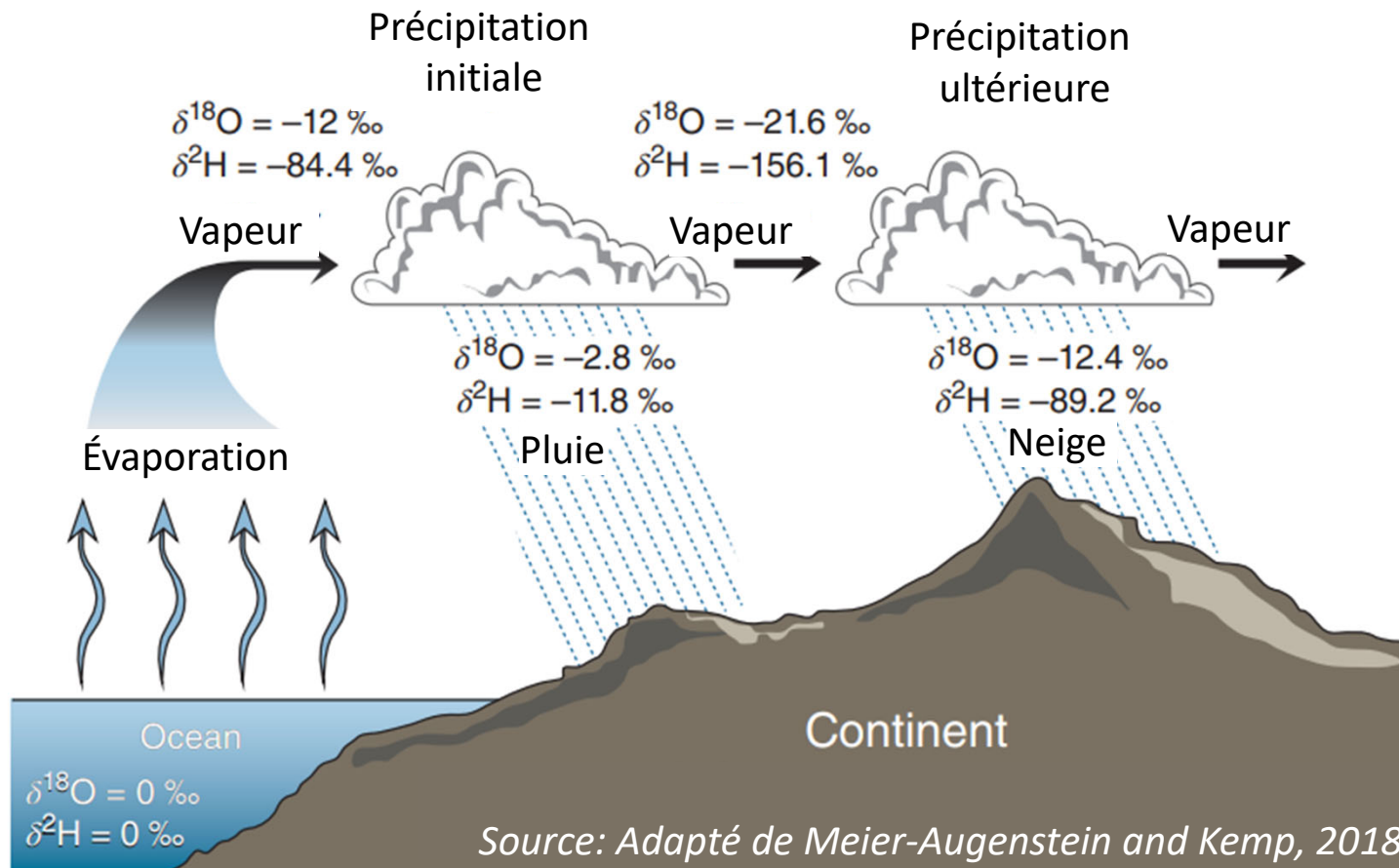
[doi: 10.7185/geochemlet.1909](https://doi.org/10.7185/geochemlet.1909)

Fractionnement isotopique: cycle de l'eau

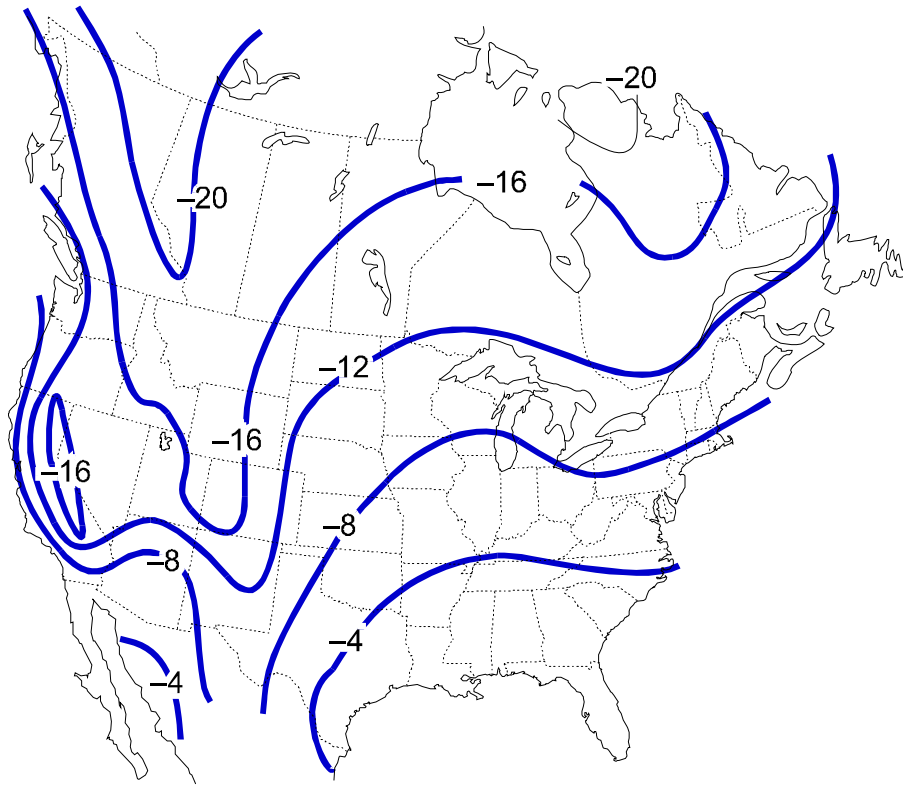


$^1\text{H}_2^{16}\text{O}$: 18 amu $^2\text{H}_2^{16}\text{O}$: 20 amu

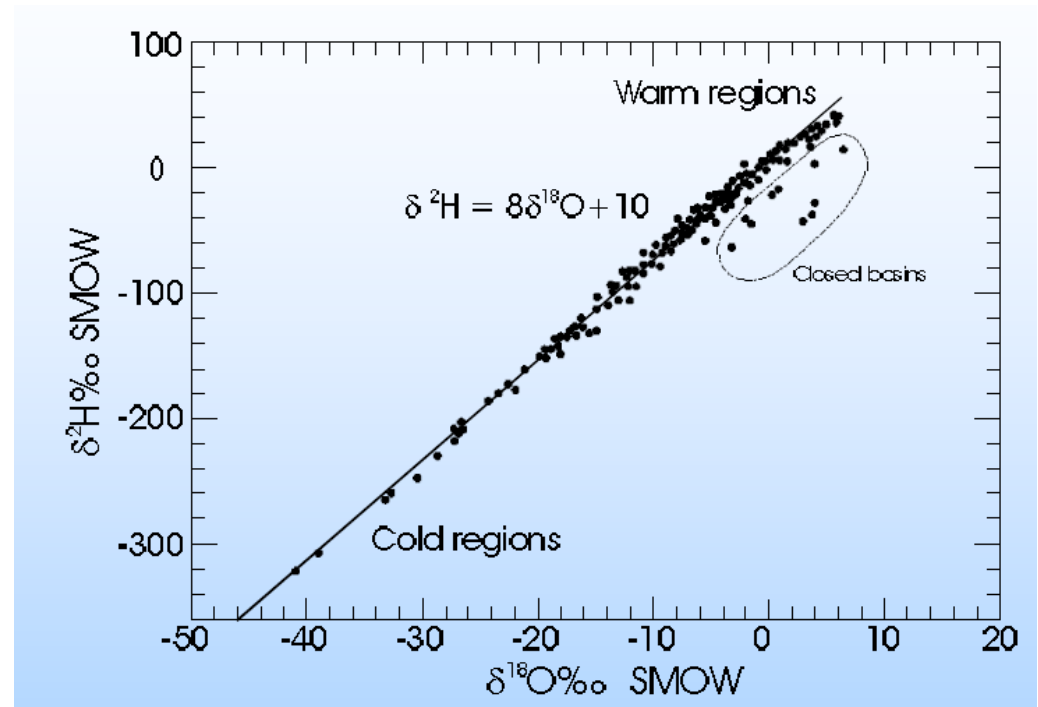
$^1\text{H}_2^{18}\text{O}$: 20 amu $^2\text{H}_2^{18}\text{O}$: 22 amu



Fractionnement d'équilibre: La condensation



Distribution du $\delta^{18}\text{O}$ dans l'eau de pluie



Fractionnement d'équilibre: La condensation

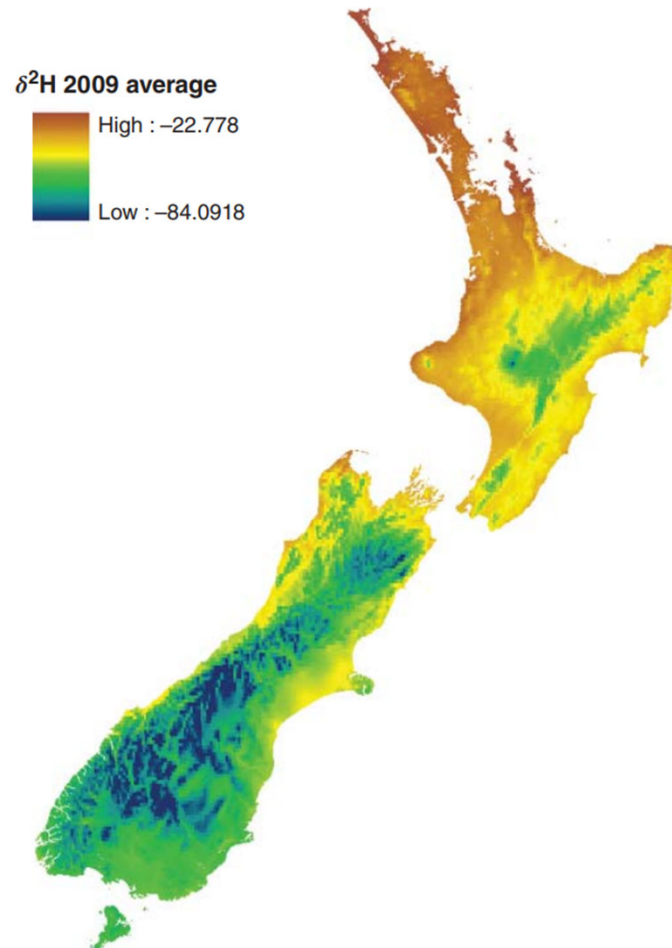
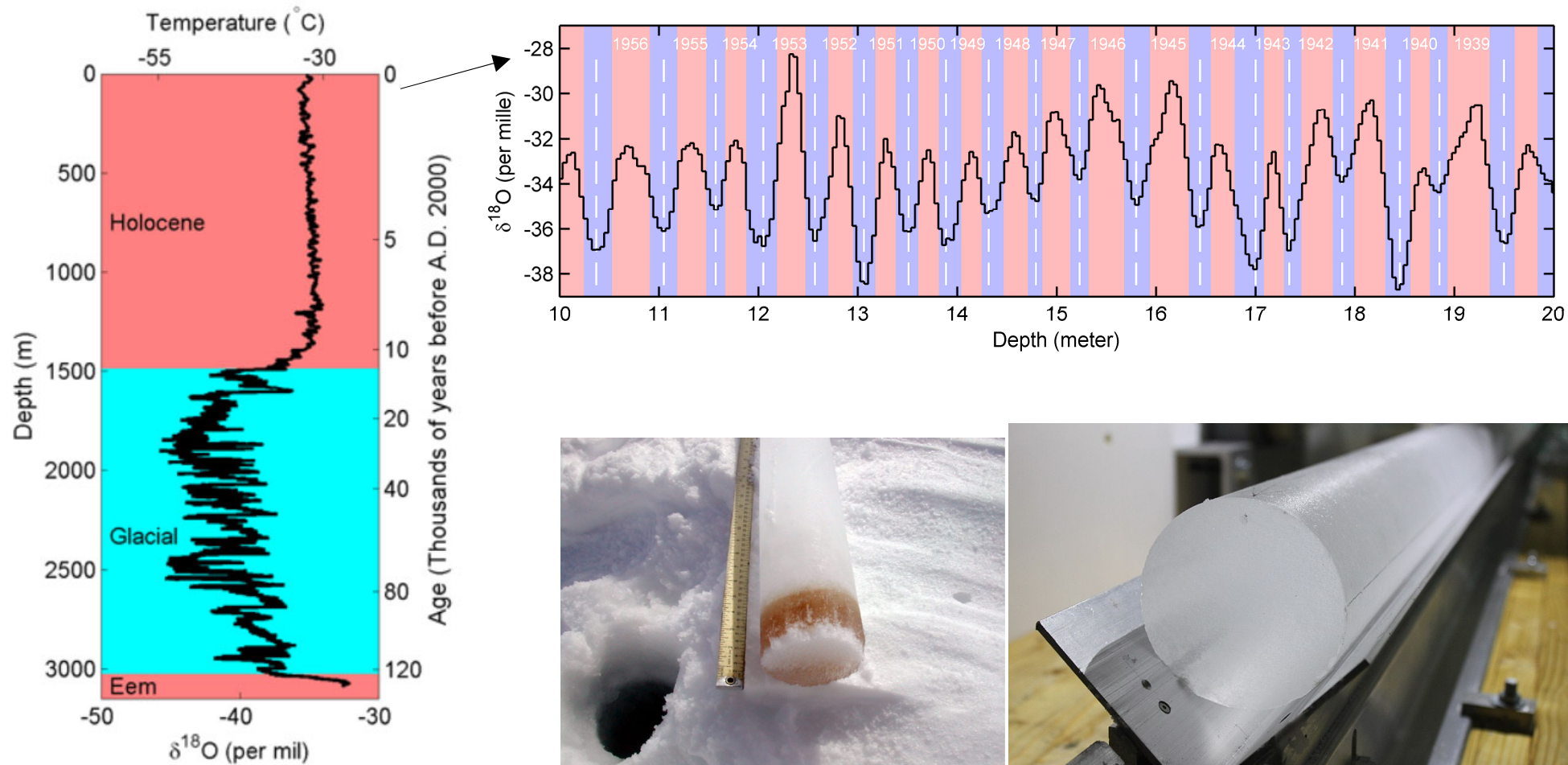


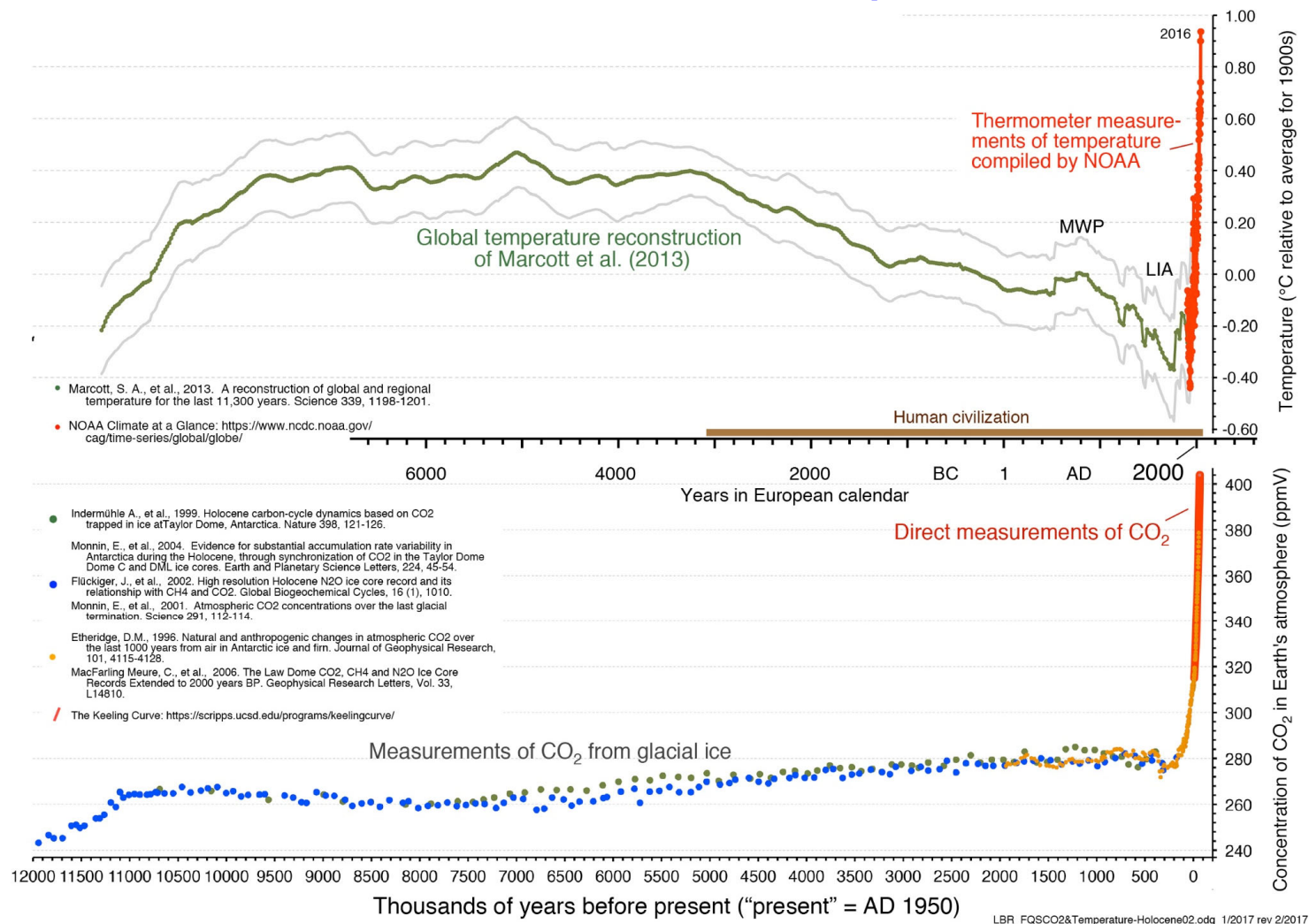
Figure I.13 $\delta^2\text{H}$ isoscape of ^2H abundance in annual precipitation of 2009 in New Zealand.
Courtesy: Reproduced with kind permission of Russell D. Frew 2015.

Fractionnement à l'équilibre: archive de glace



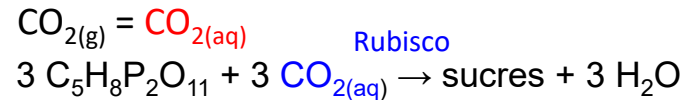
<https://www.iceandclimate.nbi.ku.dk/>

Les paléothermomètres montrent les températures de l'holocène

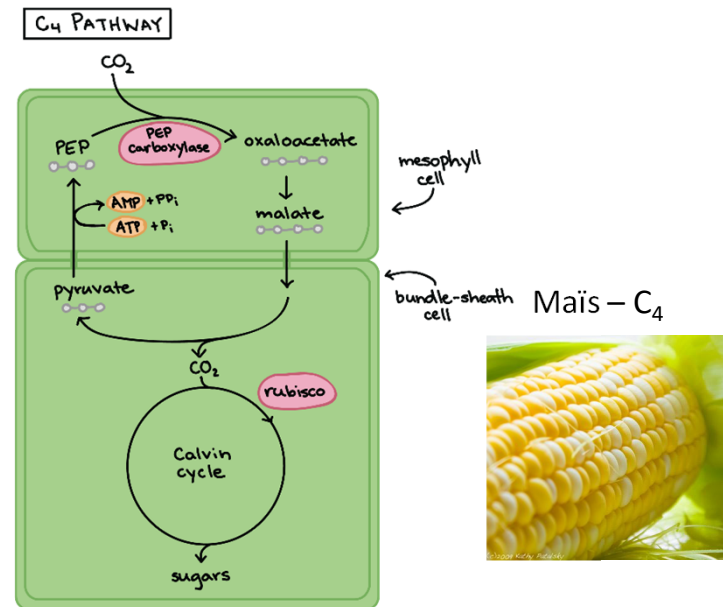
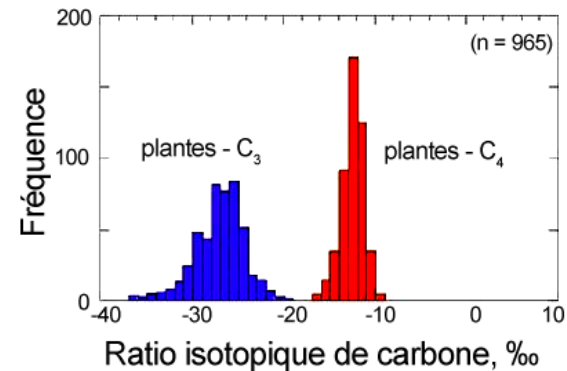
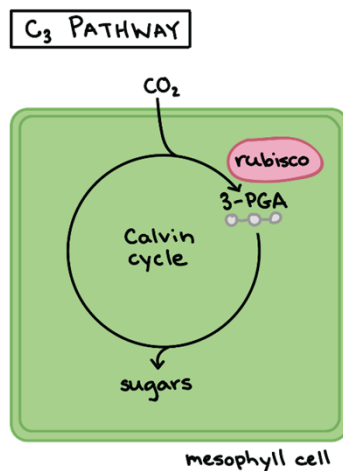
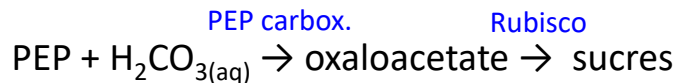
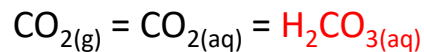


Fractionnement à l'équilibre et cinétique

- C₃ (Calvin) – Plus commun (95% des plantes)

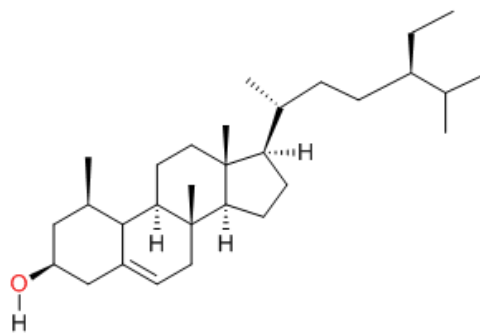


- C₄ (Hatch-Slack) – moins commun (5% des plantes, 25% du C fixé)



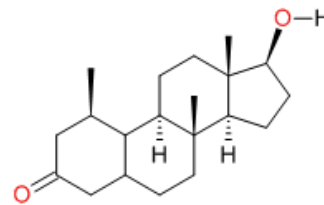
Farquhar et al. (1989) Plant Pysiol

$\delta^{13}\text{C}$ – Dopage



Phytostérol

$\delta^{13}\text{C} \geq 5\text{‰}$



Testostérone

Voie enzymatique naturelle : léger

Voie de synthèse : lourd



Floyd Landis

$\delta^{13}\text{C}$ – Produits alimentaires

La fraude alimentaire est un fléau global (OCDE, 2007) d'une valeur de plus de 200 G\$ par année :

Fraude de contenu:

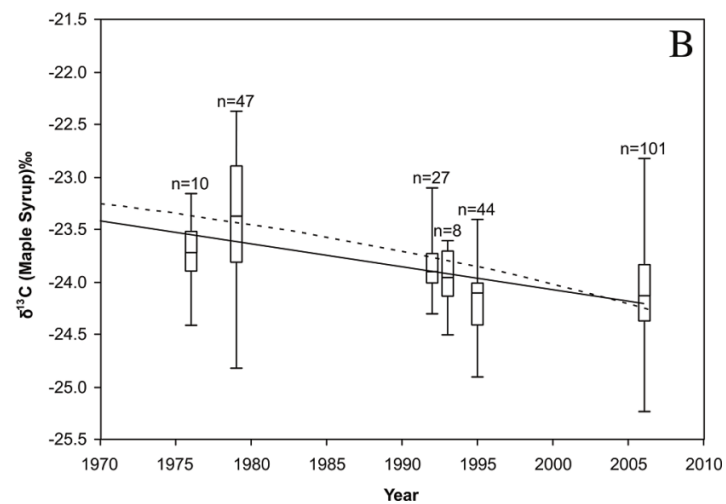
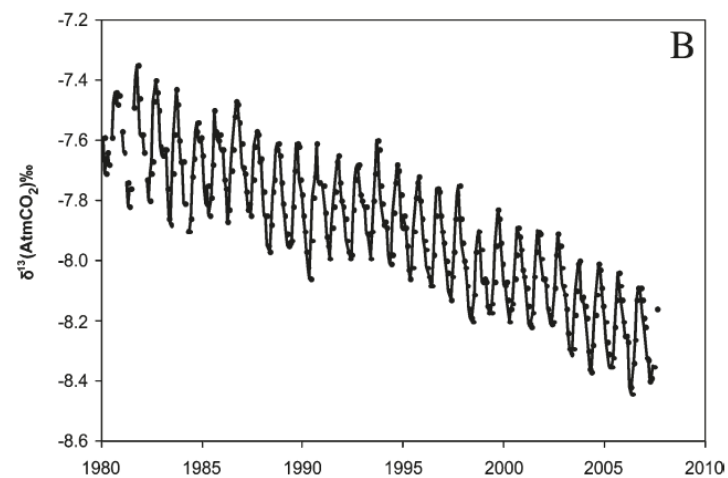
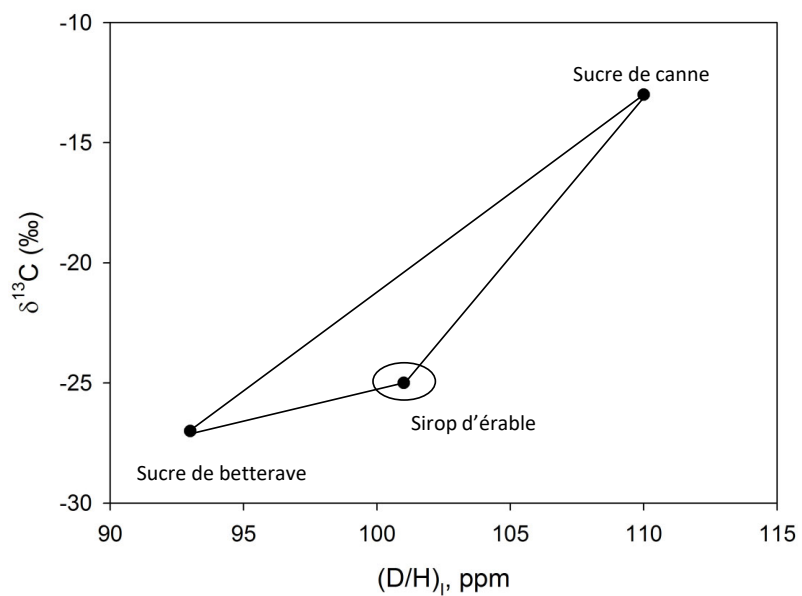
- Huile d'olive
- Huile de noix
- Sirop d'érable
- Médicaments
- Miel

Fraude de provenance:

- Vins (français et italiens)
- Whisky (écossais)
- Fragrance et parfums (parfums de fruits)
- Textile (coton égyptien)

Attention aux marques génériques sans provenance !

$\delta^{13}\text{C}$ – Sirop d'érable



G.G. Martin et al., J. Agric. Food Chem. **1996**, 44, 3206-3213
 WH Peck et SC Tubman, J. Agric. Food Chem. **2010**, 58, 2364–2367

$\delta^{15}\text{N}$ – Antiinflammatoire Naproxen

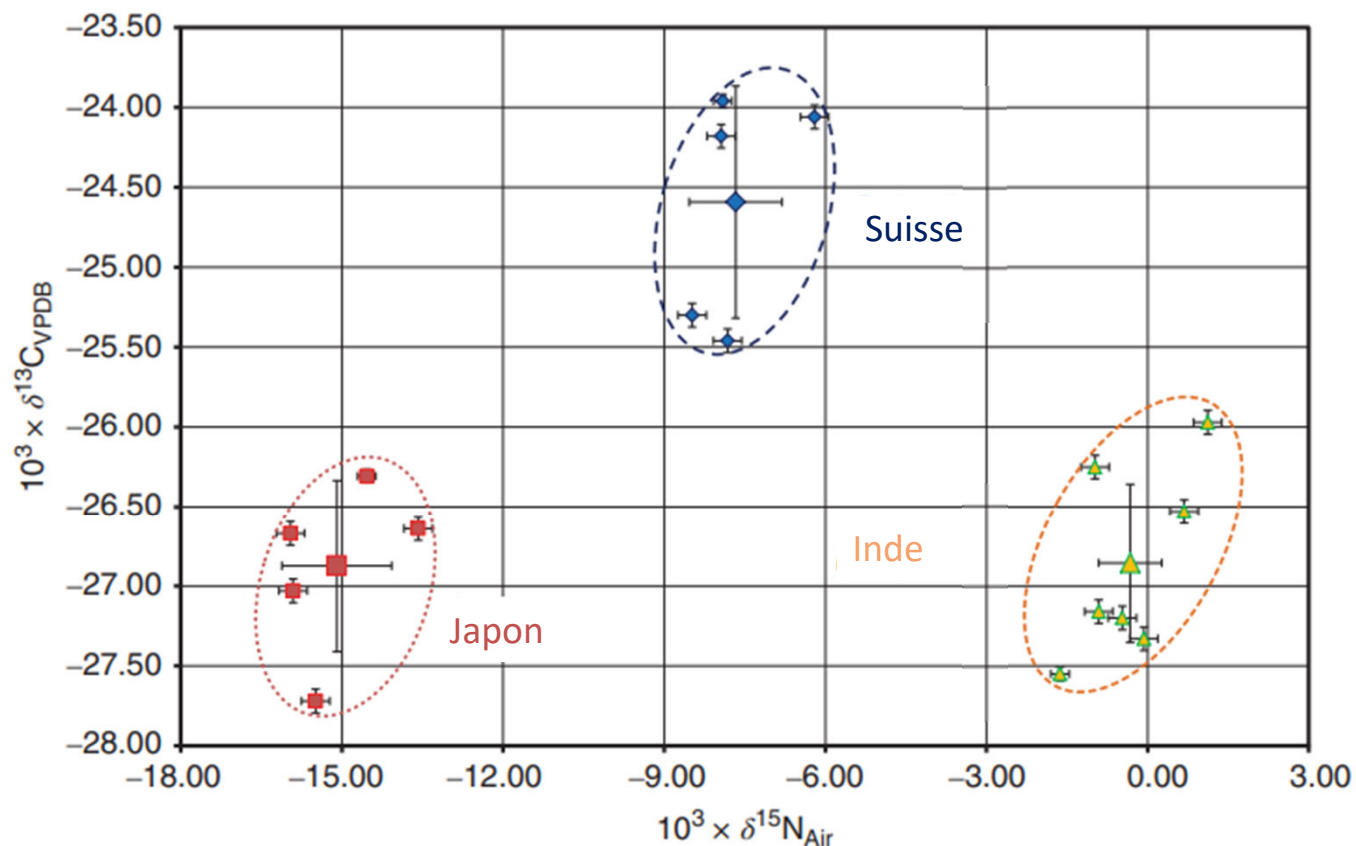
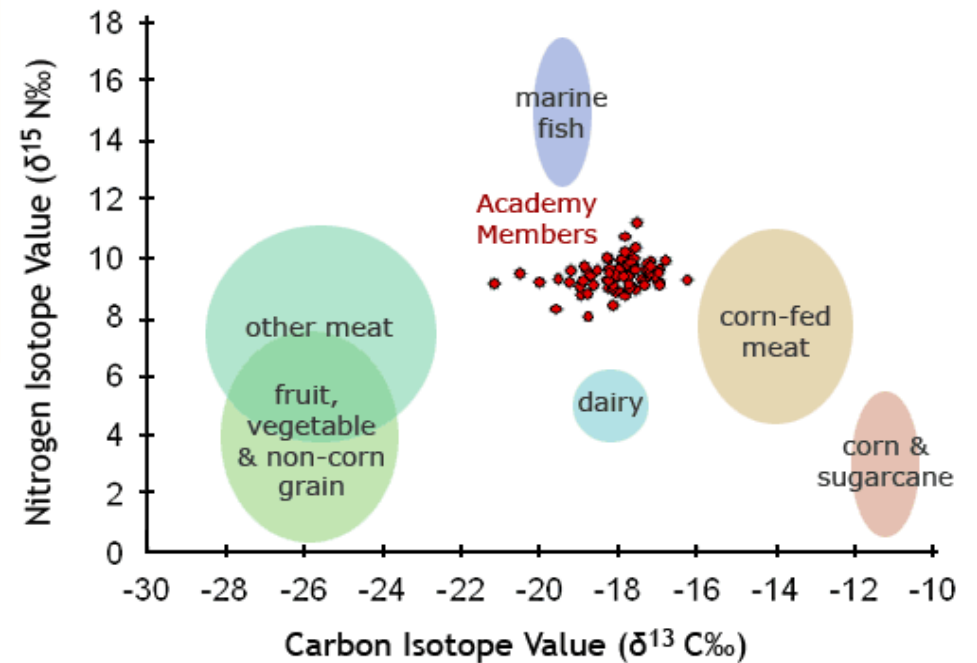
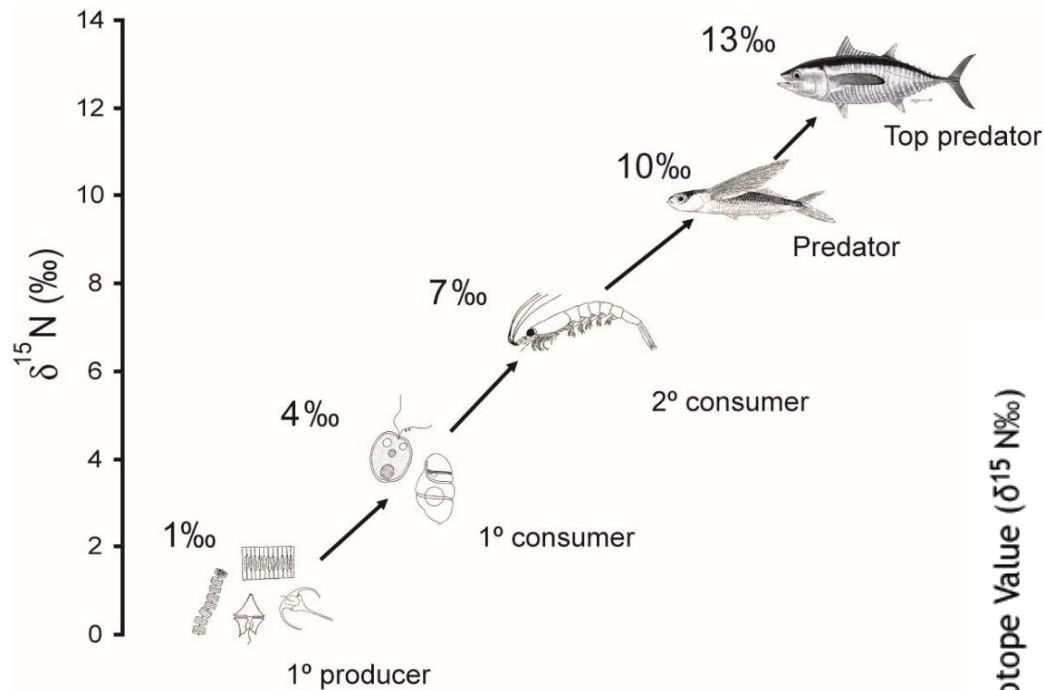
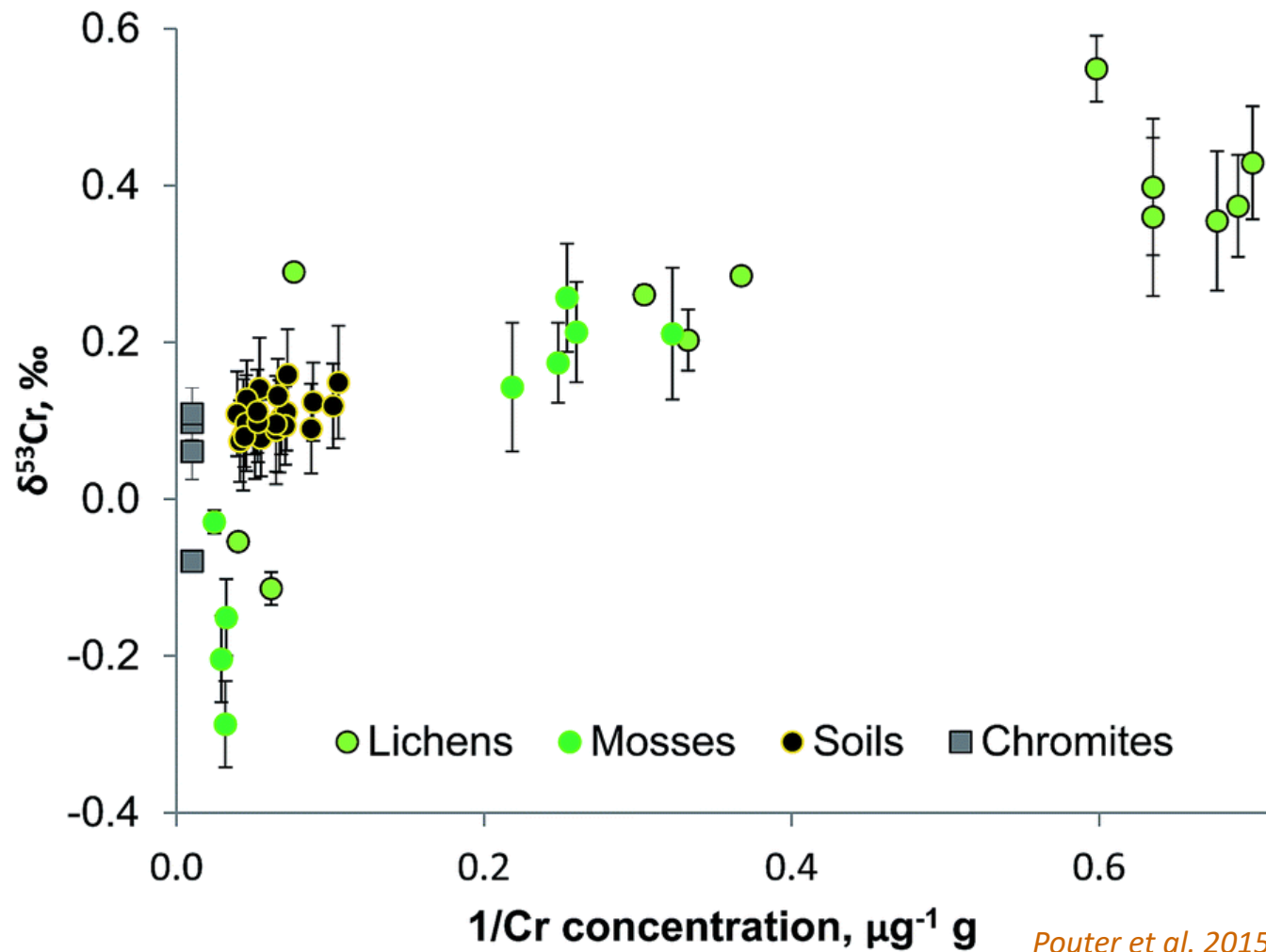


Figure I.18 An isotopic bivariate plot of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ -values of the API Folic Acid from three different manufacturers at three different locations; mean $\pm 1\sigma$ of each cluster is shown in the middle of the cluster. *Source:* Courtesy of J.P. Jasper, Nature's Fingerprint/MIT LLC, 2004.

$\delta^{15}\text{N}$ – Chaîne alimentaire (trophique)

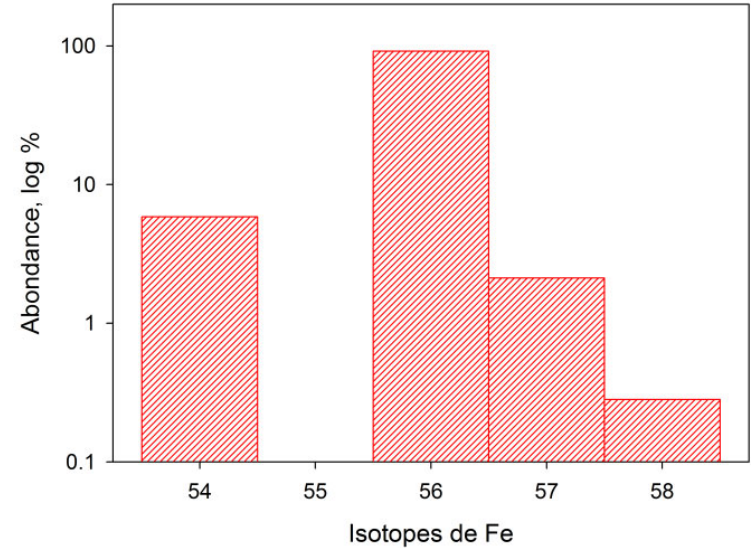
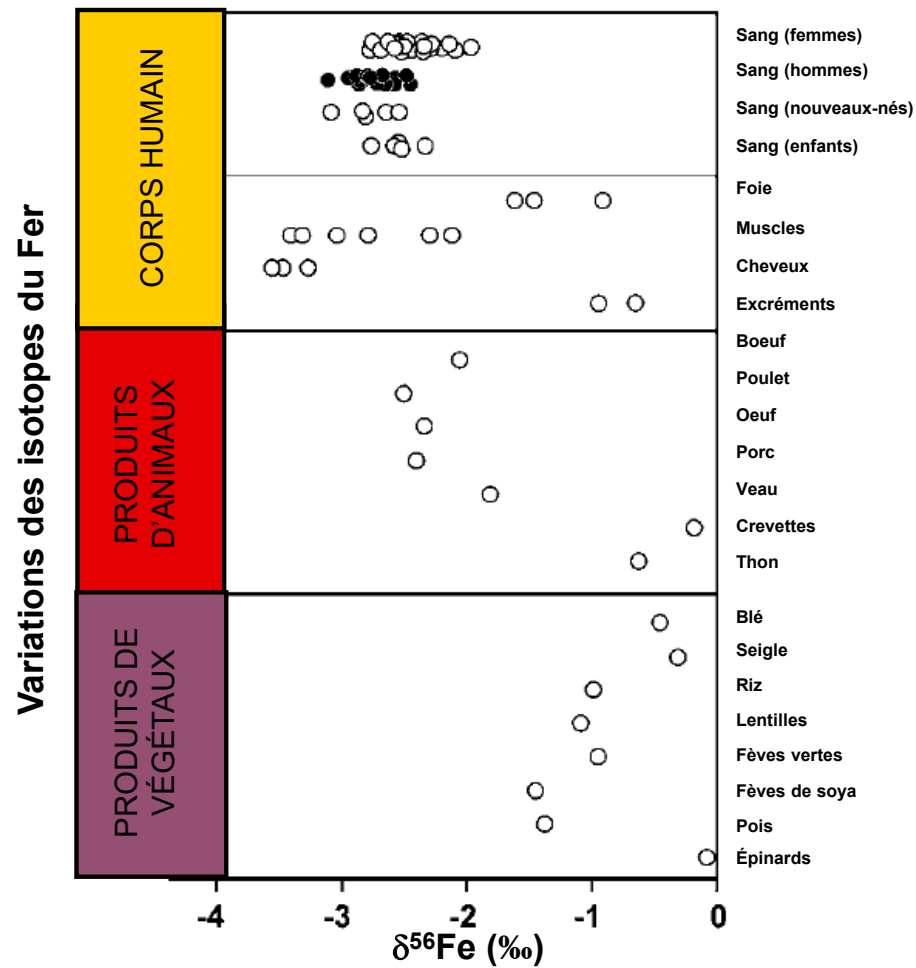


$\delta^{53}\text{Cr}$ – Pollution par le Chrome



Pouter et al. 2015 JAAS

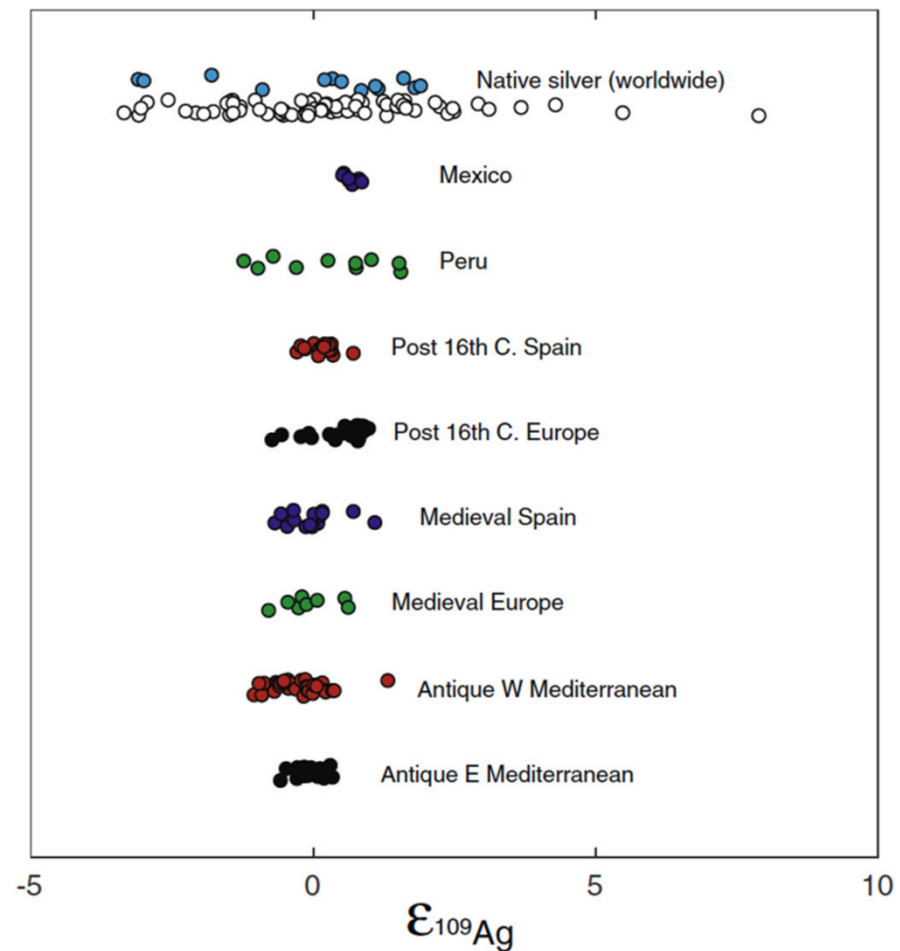
$\delta^{56}\text{Fe}$ – Métabolisme du fer



T. Walczyk et F. Von Blanckenburg, *Science*, 295 (2002) 2065

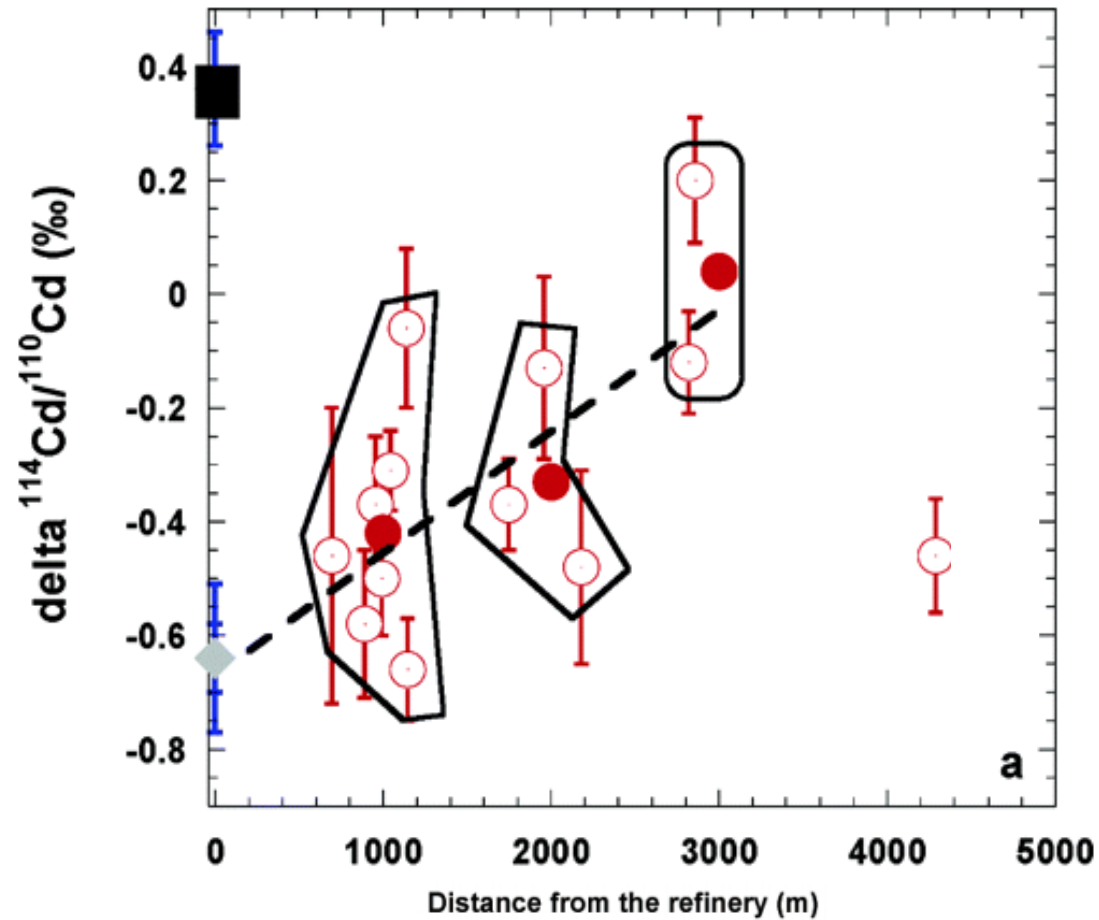
Ratio des isotopes stables de ^{109}Ag – Archéologie

$$\varepsilon^{109}\text{Ag} = \left[\frac{\left(\frac{^{109}\text{Ag}}{^{107}\text{Ag}} \right)_{\text{Sample}}}{\left(\frac{^{109}\text{Ag}}{^{107}\text{Ag}} \right)_{\text{Standard}}} - 1 \right] \times 10,000$$



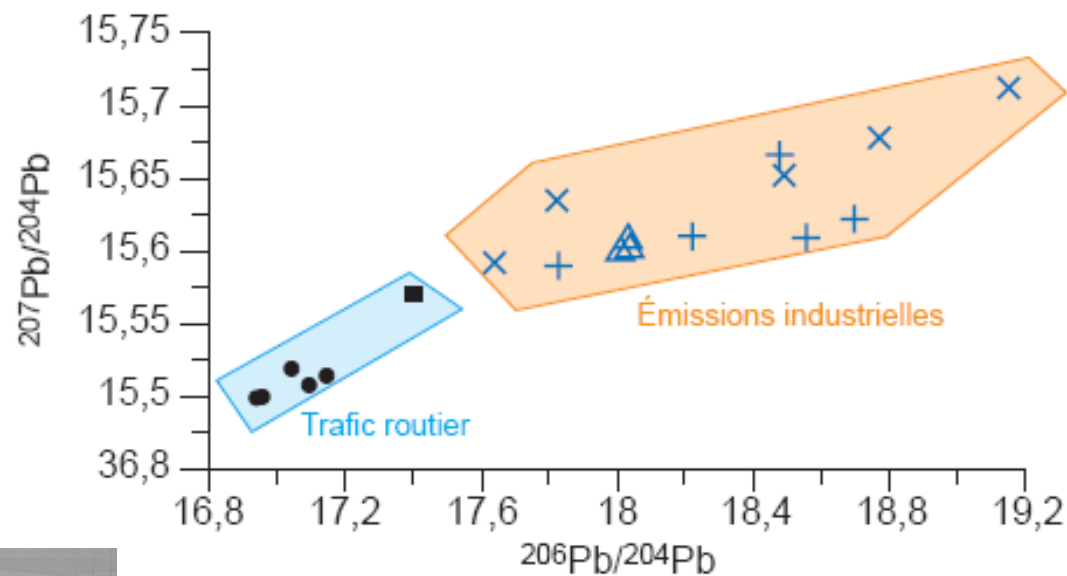
Stefen et al. (2021) J. Archeological Sci. [DOI](#)

$\delta^{114}\text{Cd}$ – Pollution par le Cadmium



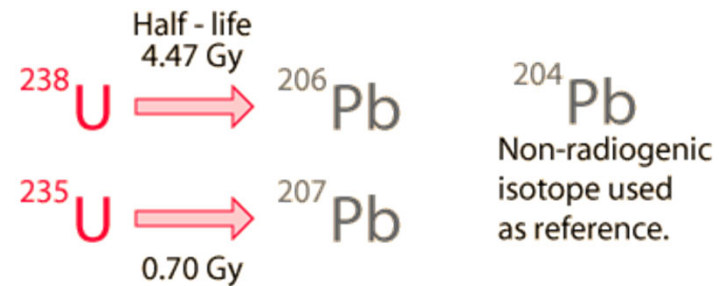
Cloquet et al. 2006 Environ. Sci. Technol.

Ratio des isotopes stables du Pb – Sources de pollution

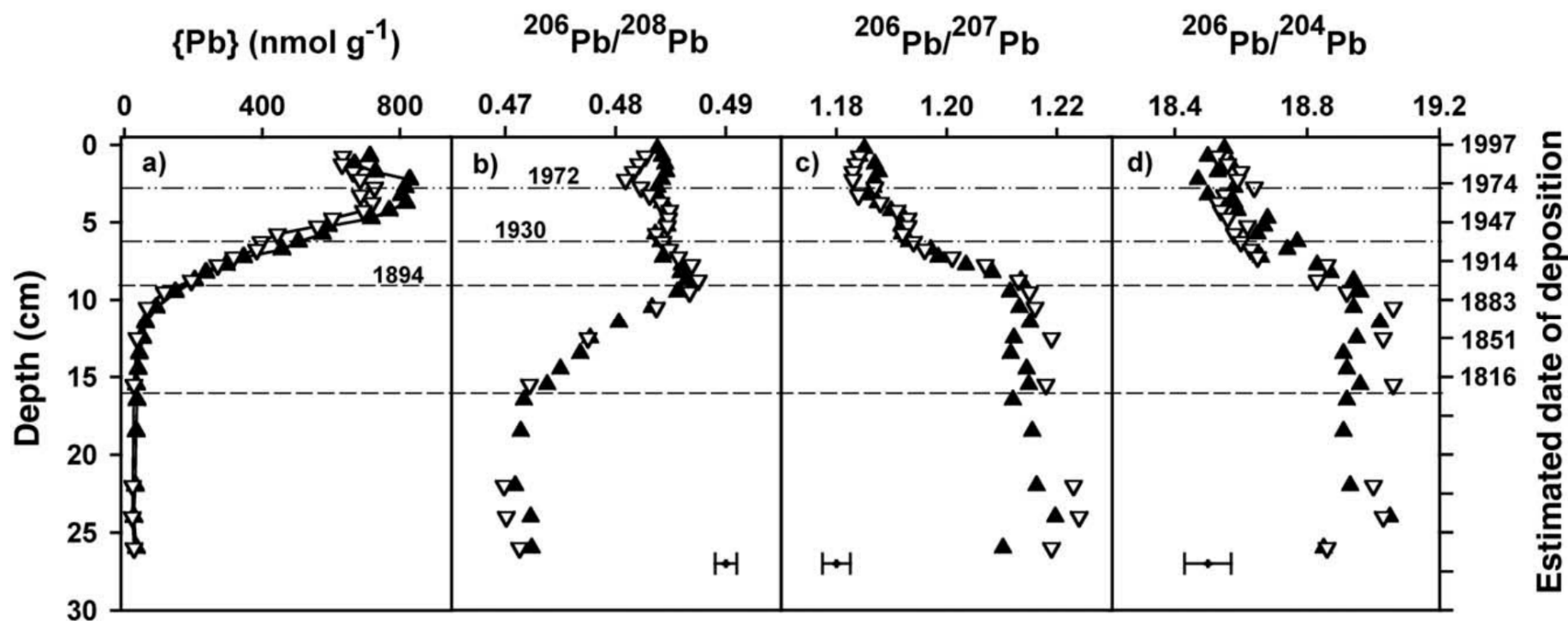


Compositions isotopiques du plomb des sources de pollution dans l'atmosphère parisienne.

Discrimination des sources de Pb industrielles



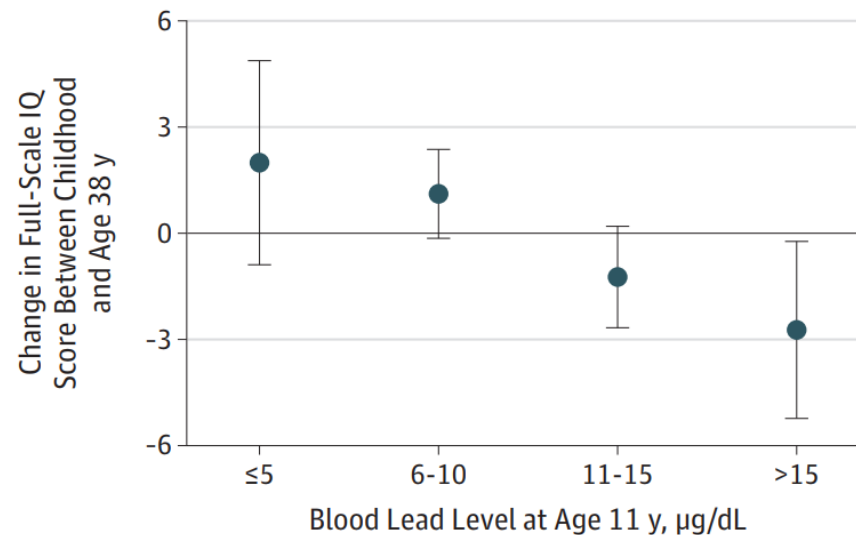
Concentration et signature du Pb, région de Québec



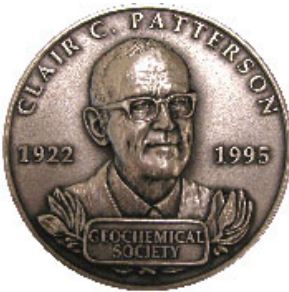
Gallon et al. (2005) *Geochim. Cosmochim. Acta*

Déclin cognitif causé par le Pb

A Change in full-scale IQ by childhood blood lead levels



Reuben et al. JAMA. 2017

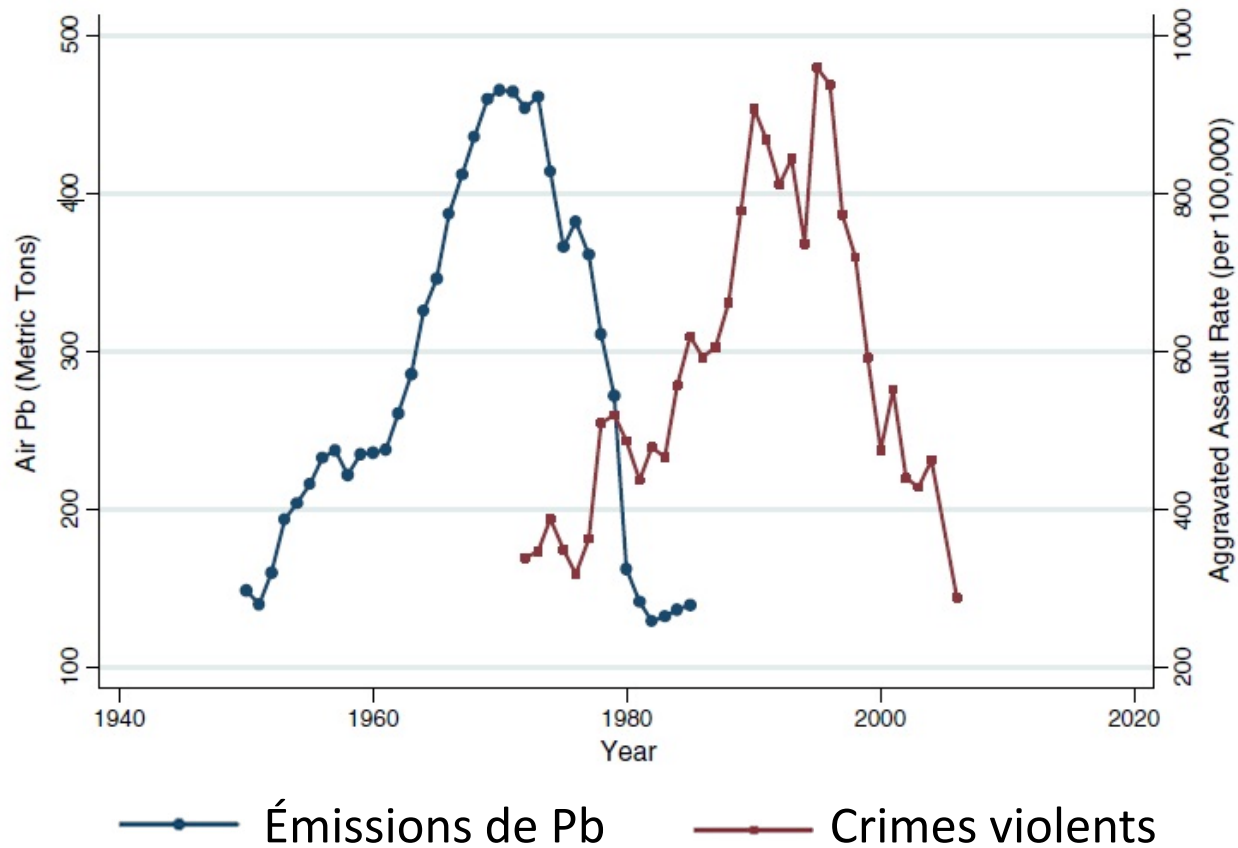


*“Contaminated and Natural
Lead Environments of Man”
1965*

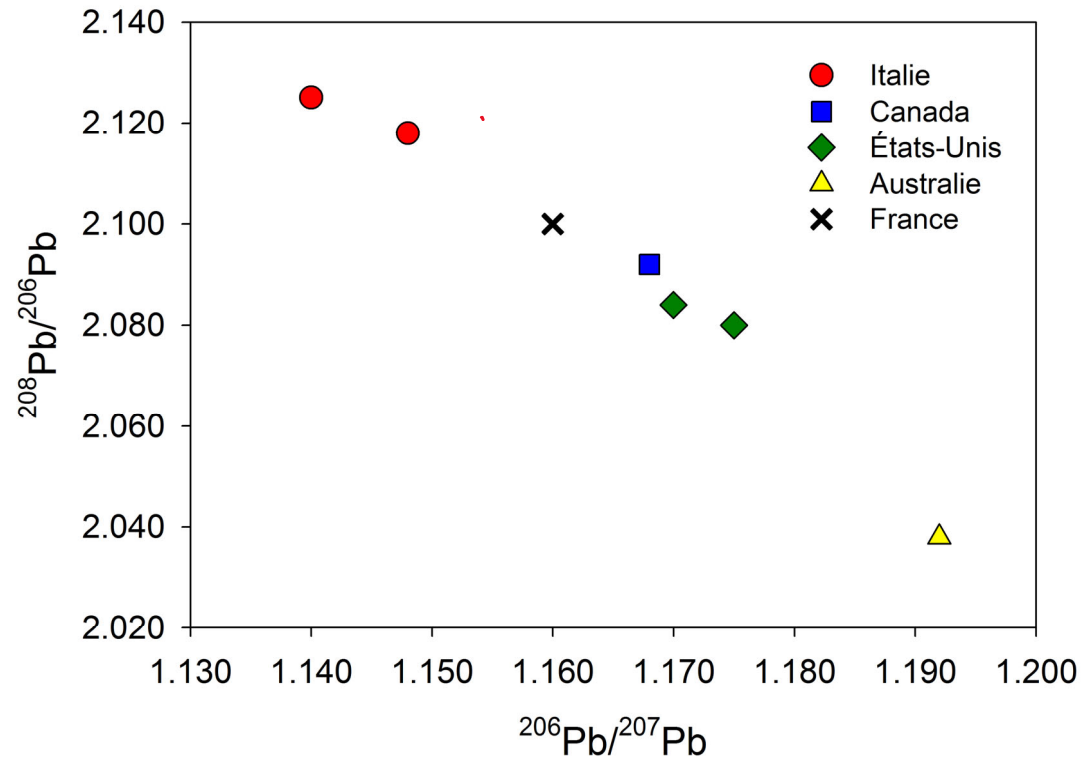
Clair Patterson Award

Suivi environnemental et épidémiologique: Plomb

H.W. Mielke, S. Zahran / Environment International 43 (2012) 48–55

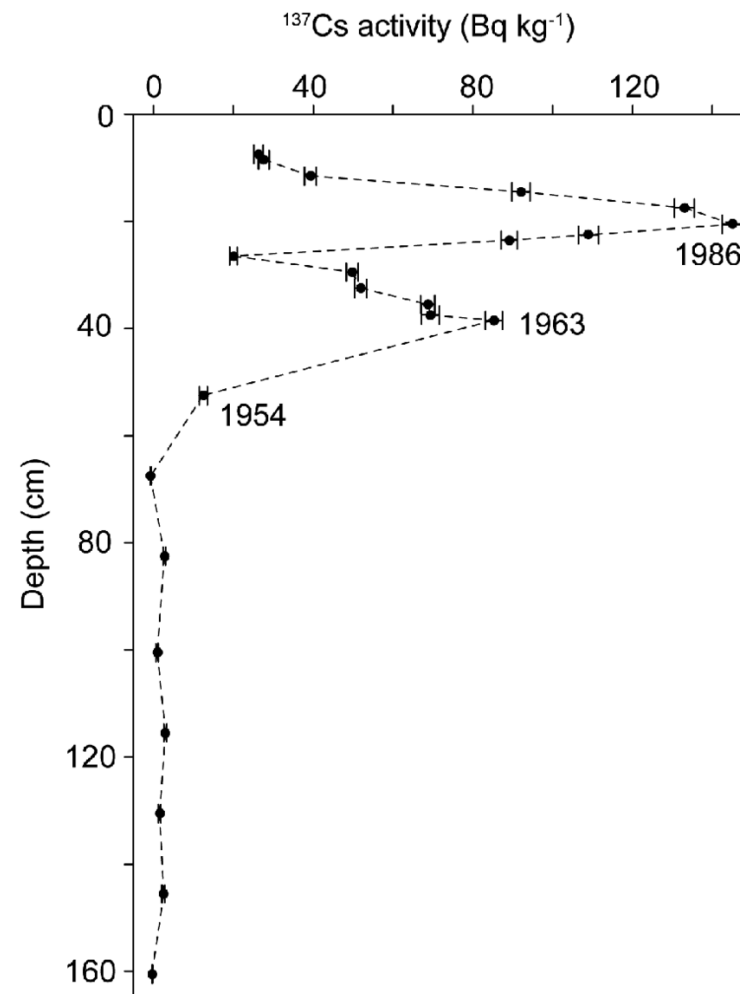


Ratio Pb – Contrefaçon du vin

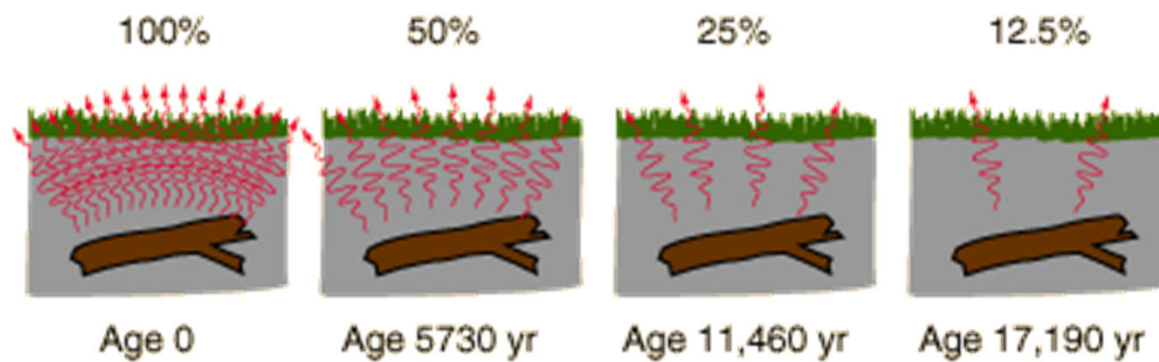
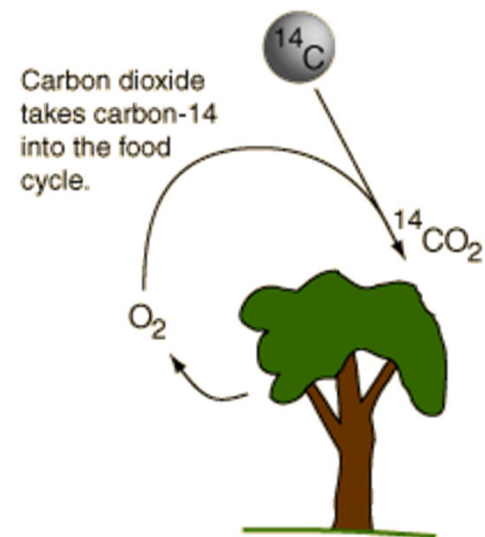
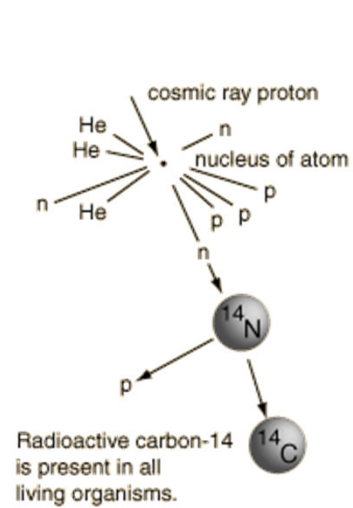


Ratios isotopiques du plomb dans le vin rouge de 5 pays.

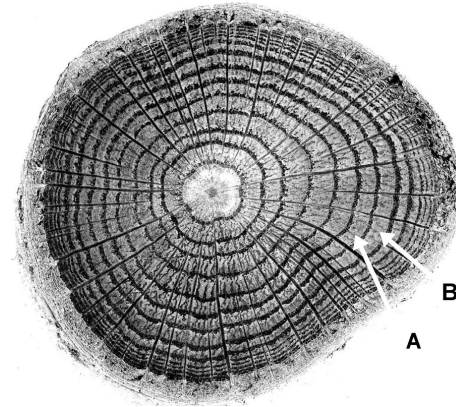
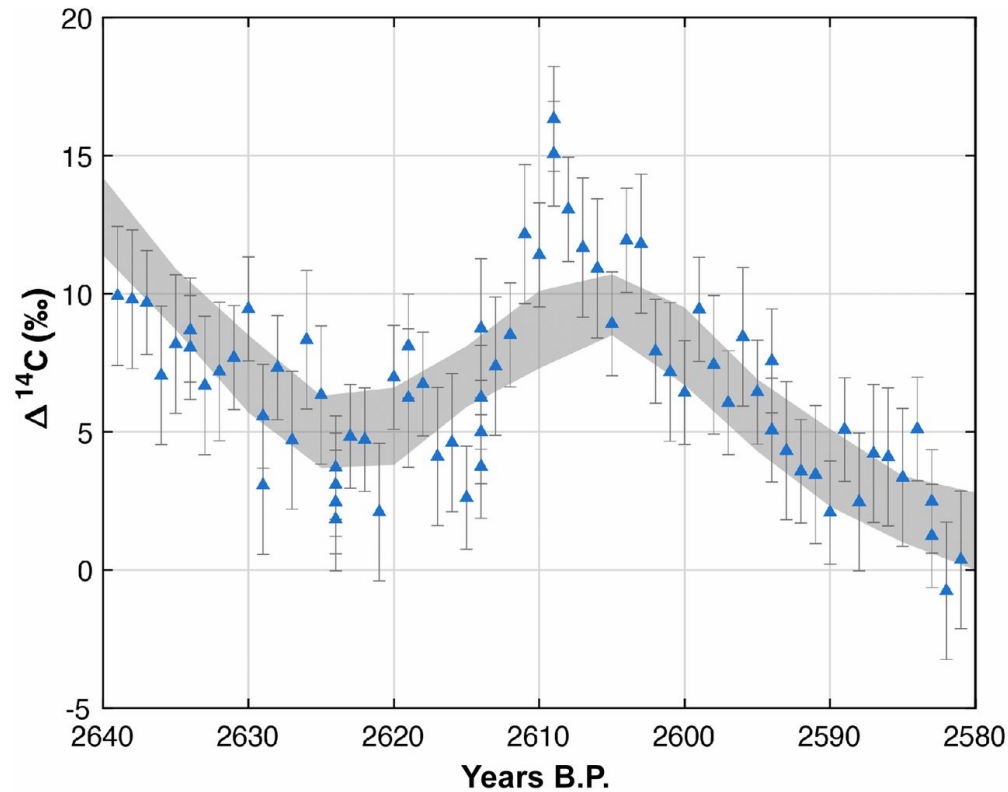
Suivi environnemental : Radiocésium (^{137}Cs)



Datation au ^{14}C



Reconstruction du flux de neutrons solaire



Stuiver et Braziunas (1989) Atmospheric ^{14}C and century-scale solar oscillations, *Nature*

Applications pratiques des isotopes

La signature isotopique est un indice:

Des processus (bio)chimiques (isotopes **légers**)

De l'âge (isotopes **radioactifs**)

Des sources (isotopes **lourds**)

Applications pratiques des isotopes

H & O: météorologie

O: paléothermométrie

C,N : niveau trophique

Si: géothermometrie

Fe/S: géo/biogéochimie, processus

Cr/Pb: enquêtes environnementales

Géochronologie:

^{210}Pb : < 150 a

^{14}C : < 60 000 a

U/Th: < 1 million a

Sm/Nd: < 2 milliards a

Pb/Pb: < 3 milliards a